Sulabham

APPSC AEE 2026 · Module 4: Building Materials

Paper 3 Civil Engineering · Post Codes 1, 3, 4 and 7 · Study week 8 · English medium

How to use this module

1. **Understand:** read the topics in order — basic idea → why → how → analogy → traps.
2. **Connect:** study the photo and the diagram, then the "Connections" line.
3. **Remember:** learn the memory code, close the page and answer the self-recall questions aloud.
4. **Apply:** take the 5-question quiz at the end of each topic. Tap an option; the answer and the reason for every option appear only after you choose.
5. At the end, write the Notebook page from memory, then take the Module 4 test.

Week plan (Week 8): Day 1 → Topics 1–2 · Day 2 → Topics 3–4 · Day 3 → Topics 5–6 · Day 4 → Topic 7 · Day 5 → Topic 8 · Day 6 → Topics 9–10 · Day 7 → Notebook page + test. This is a memory-heavy module: values and names matter, so use the memory codes and the Notebook page.

How Building Materials was asked in APPSC AEE 2023 (Paper III Civil, 22 Aug 2023, Shift 2)

Evidence (counted from the official paper and key)
Building Materials had 10 of the 150 questions (Q10 to Q19). Only this one shift was available to us, so topics with a zero below were simply not picked in this paper.

Topic	Questions	Share
Topic 1: Timber — structure, moisture, strength	2	
Topic 2: Timber — defects, seasoning, plywood	1	
Topic 3: Bricks — earth, defects, classification	2	
Topic 4: Masonry strength; building stones	1	
Topic 5: Cement — compounds and types	0	
Topic 6: Cement — tests	0	
Topic 7: Mortar and lime	0	
Topic 8: Concrete — ingredients, w/c ratio	1	
Topic 9: Fresh and hardened concrete, NDT	1	
Topic 10: Mix design; steel	2	

How the 10 questions were framed	Count	Question numbers
Direct one-line fact or definition	6	Q11, Q12, Q14, Q16, Q18, Q19
Match the lists	2	Q10, Q13
Statement with two conclusions	1	Q15
Two statements (S1, S2)	1	Q17
Numerical	0	—

Our analysis (not a prediction)
All ten questions were factual — no calculation. Three of them (Q13 on stones, Q16 and Q17 on steel) came from materials that the syllabus line does not name, so this module adds stones to Topic 4 and steel to Topic 10. Two questions quoted an Indian Standard by number (IS 1786, IS 11215), so learn the code number with the fact. Two were again built by reversing a true statement (Q15 shrinkage direction, Q17 ductility).

Topic 1: Timber — Structure, Species, Moisture and Strength

Syllabus line: *Timber: different types and species of structural timber, density–moisture relationship, strength in different directions.*

Asked in APPSC AEE 2023, Paper III Civil, 22 Aug Shift 2 (verified with official key): 2 questions

- Statement with two conclusions on the **fibre saturation point**: above it, the properties of wood do not change with moisture (conclusion 1, true); the paper then claimed shrinkage is greatest in the **longitudinal** direction (false — it is the least). Key: only conclusion 1 (Q15).
- Methods of finding the moisture content of wood under IS 11215: oven drying, distillation and electrical moisture meter are valid; the **'torsion balance moisture meter'** is not a method (Q19).

1. Basic idea

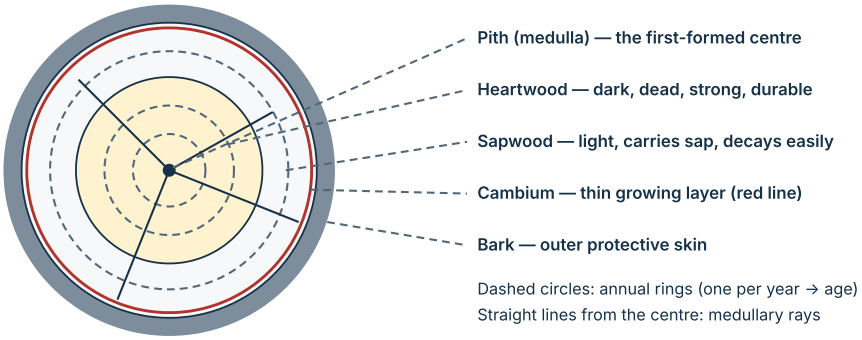
Timber is wood that is fit for engineering use. A **standing** tree gives standing timber; a felled trunk is **rough** timber (a log); sawn to market sizes it is **converted** timber. Wood is made of long hollow cells (fibres) running along the trunk, like a tight bundle of drinking straws. Almost every property of timber follows from this one fact.

2. Why it matters

Timber is light, easily worked, strong for its weight and a good insulator, so it is used for doors, formwork, scaffolding, roof trusses, piles and sleepers. Its weakness is that it changes with **moisture** and is attacked by fungi and insects — which is what the exam asks about.

3. How it works

Basis	Types	Examples
Mode of growth	**Exogenous** — grow outward, adding one ring a year (used in engineering). **Endogenous** — grow inward, fibrous	Exogenous: teak, sal, deodar. Endogenous: bamboo, palm, cane
Leaf type (exogenous)	**Softwood** — conifers, needle leaves, light, resinous, distinct rings, strong in tension. **Hardwood** — broad leaves, dense, dark, strong in compression and shear too	Softwood: deodar, chir, pine, fir. Hardwood: teak, sal, shisham, oak, babul



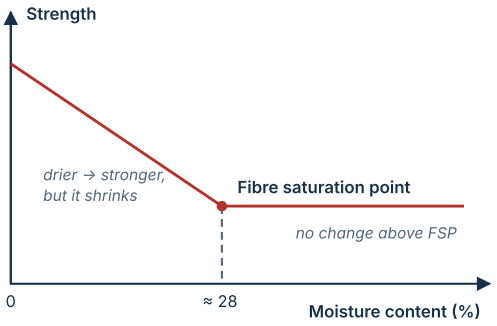
Structural timber is cut from the heartwood. Count the annual rings to find the age of the tree; the medullary rays hold the rings together.

Species	Character	Main use
Teak	Hard, durable, dimensionally stable, resists white ants and dry rot	Best all-round structural and joinery timber — doors, furniture, ship building
Sal	Very hard, heavy, durable in the ground and under water	Railway sleepers, bridges, piles, beams
Deodar	Softwood, light, oily, durable	Sleepers and structural work in the hills
Shisham (sissoo)	Hard, takes fine polish	High-class furniture, plywood
Babul	Very hard, tough	Tool handles, cart wheels, agricultural tools
Mango	Soft, cheap, not durable	Packing cases, cheap shuttering and furniture
Bamboo	Endogenous, flexible, strong in tension	Scaffolding, thatched roofs
Chir / pine, fir	Softwood, resinous	Packing cases, shuttering, temporary works

IS 883 places structural timbers in three groups by modulus of elasticity: **Group A** — E above 12.6 kN/mm²; **Group B** — 9.8 to 12.6 kN/mm²; **Group C** — 5.6 to 9.8 kN/mm².

Moisture in wood

- **Moisture content** = (wet weight − oven-dry weight) ÷ oven-dry weight × 100. It is a percentage of the **dry** weight, so green timber can have more than 100 %.
- Water is held in two ways: **free water** inside the cell cavities, and **bound water** soaked into the cell walls.
- **Fibre saturation point (FSP)**: the cell walls are fully saturated but the cavities hold no free water — about **25 to 30 %**.
- **Above FSP** only free water changes, so strength and size stay constant. **Below FSP** the walls dry: the wood **shrinks** and its **strength rises**.
- **Shrinkage:** tangential (along the rings) is the largest, radial about half of it, **longitudinal the least** (almost nil).
- **Density** rises with moisture, so it is quoted at a standard 12 % moisture. Denser timber is generally stronger.
- **Methods of finding moisture (IS 11215):** oven-drying (standard), distillation (for oily or treated timber), electrical moisture meter (quick, in the field).



Strength in different directions

Tension along the grain: highest
Compression along the grain: about half of that
Compression across the grain: much lower
Shear along the grain: low (fibres slide)
Tension across the grain: lowest (fibres split)

Timber is anisotropic: its properties depend on the direction of the load relative to the grain.

Below the fibre saturation point strength rises as the wood dries; above it nothing changes. This graph is the answer to 2023 Q15.

4. Everyday analogy

Think of a bundle of drinking straws tied together. Pull or push **along** the straws and the bundle is strong. Squeeze it from the side and the straws flatten; try to pull them apart sideways and they simply separate. That is why timber is strong along the grain and weak across it. A wet sponge works for moisture: squeezing out the loose water (free water) does not change its size, but once the sponge material itself dries (bound water) it shrinks and stiffens.

5. Confusing areas and exam traps

- **Softwood and hardwood are botanical terms**, not a measure of hardness. Balsa is a hardwood.
- Shrinkage order: **tangential > radial > longitudinal**. The 2023 paper reversed it.
- Strength changes only **below** the FSP. Above it, wetter wood is heavier but not weaker.
- Heartwood gives structural timber; sapwood is weaker and decays first.
- Moisture content is on the oven-dry weight, not the wet weight.

6. Connections

Shrinkage below FSP is the cause of the seasoning defects in Topic 2. Anisotropy explains why plywood (Topic 2) crosses the grain of alternate layers. Moisture content on dry weight is defined the same way for soils (Soil Mechanics module).

Memory code: "T-R-L" and "Dry below, steady above"

T-R-L	Shrinkage falls in this order: Tangential, Radial, Longitudinal
Dry below	Below FSP (about 25–30 %): shrinks and gets stronger
Steady above	Above FSP: size and strength do not change
O-D-E	Moisture methods in IS 11215: Oven-drying, Distillation, Electrical meter

Self-recall (close the notes, answer aloud, then check)

1. What is the fibre saturation point and its usual value?
   The moisture content at which the cell walls are saturated but the cell cavities are empty of free water; about 25 to 30 %.
2. In which direction is shrinkage the greatest and the least?
   Greatest tangentially (along the annual rings); least longitudinally (along the grain).
3. Name two softwoods and two hardwoods used in India.

Softwoods: deodar, chir (pine). Hardwoods: teak, sal (also shisham, babul).

Quick check: 5 questions

1. A sample of timber weighs 120 g when cut and 100 g after oven drying. Its moisture content is

   1. 16.7 %
   2. 120 %
   3. 83.3 %
   4. 20 %

1. Divides by the wet weight; moisture content is on the oven-dry weight.
2. This is wet weight ÷ dry weight.
3. This is dry weight ÷ wet weight.
4. Correct. (120 − 100)/100 × 100 = 20 %.

2. **Statement:** A timber beam is dried from 40 % moisture content to 30 %, which is still above its fibre saturation point.
Conclusion 1: Its weight decreases.
Conclusion 2: Its strength increases appreciably.

   1. Only conclusion 1 is correct
   2. Only conclusion 2 is correct
   3. Both conclusions 1 and 2 are correct
   4. Neither conclusion 1 nor 2 is correct

1. Correct. Free water leaves, so the beam is lighter, but strength changes only below the fibre saturation point.
2. Conclusion 2 is false above the FSP.
3. Conclusion 2 is false.
4. Conclusion 1 is true: losing free water reduces weight.

3. Match List-I with List-II:

List-I (timber)	List-II (typical use)
A. Sal	1. Scaffolding
B. Bamboo	2. High-class furniture
C. Babul	3. Railway sleepers and piles
D. Shisham	4. Tool handles and cart wheels

   1. A-3, B-1, C-2, D-4
   2. A-1, B-3, C-4, D-2
   3. A-3, B-1, C-4, D-2
   4. A-4, B-1, C-3, D-2

1. Babul goes with tool handles, shisham with furniture.
2. Sal is the sleeper timber; bamboo is used for scaffolding.
3. Correct. Sal → sleepers and piles; bamboo → scaffolding; babul → tool handles; shisham → furniture.
4. Sal is not used for tool handles.

4. Consider the following statements:
   I. Softwoods come from coniferous trees with needle-shaped leaves.
   II. Heartwood is stronger and more durable than sapwood.
   III. The age of an exogenous tree can be found from its annual rings.
   IV. Timber is strongest in tension across the grain.

   Which of the above statements are correct?

   1. I and II only
   2. I, II and III only
   3. II, III and IV only
   4. I, II, III and IV

1. Leaves out III, which is true.
2. Correct. IV is false: tension across the grain is the weakest direction; along the grain is the strongest.
3. IV is false.
4. Includes the false statement IV.

5. Which of the following is NOT a method given in IS 11215 for finding the moisture content of timber?

1. Oven-drying method
2. Distillation method
3. Electrical moisture meter method
4. Torsion balance moisture meter method

1. This is the standard method.
2. Used for oily or preservative-treated timber.
3. The quick field method.
4. Correct. It is not a method in the code (2023 Q19).

Topic 2: Timber — Defects, Seasoning, Preservation and Plywood

Syllabus line: *Timber: defects, preservations, and plywood.*

Asked in APPSC AEE 2023, Paper III Civil, 22 Aug Shift 2 (verified with official key): 1 question

- Identify the defect from its description — radial splits **wide at the circumference and narrowing towards the centre**, caused by severe frost or fierce sun: **star shake** (Q14).

1. Basic idea

A **defect** is anything that lowers the strength, durability or appearance of timber. Defects come from five sources: the growing tree, drying (seasoning), sawing (conversion), fungi and insects. **Seasoning** is controlled drying of timber. **Preservation** is treatment with chemicals so that fungi and insects cannot live on it.

2. Why it matters

A freshly felled tree may hold more than half its weight as water. If it is used as it is, it shrinks, warps and rots in the structure. Seasoning and preservation turn a perishable material into a durable one.

3. How it works



Heart shake
from the pith outward, widest at the centre

Star shake
from the bark inward, widest at the circumference

Cup shake
curved split along an annual ring (part of it)

Three shakes that are easily confused. Remember where the crack is widest. A cup shake that runs round the whole ring is a ring shake.

Source	Defects	One-line identification
Natural (in the growing tree)	Knots, shakes, rind gall, twisted fibres, upsets, burls, druxiness, foxiness, dead wood	**Knot** — base of a branch embedded in the wood. **Rind gall** — curved swelling where a branch was badly cut off. **Upset (rupture)** — fibres crushed by felling or wind. **Druxiness** — white decayed spots. **Foxiness** — red or yellow tinge from poor ventilation or over-maturity
Seasoning	Bow, cup, twist, spring (all called warp), check, split, honeycombing, case hardening, collapse	**Bow** — curved along the length. **Cup** — curved across the width. **Check** — crack separating fibres, not through the piece. **Split** — a check that goes through. **Case hardening** — surface dries fast and sets, the inside is still wet. **Honeycombing** — internal cracks
Conversion (sawing)	Chip mark, torn grain, diagonal grain, wane	**Wane** — the original rounded surface left on a sawn piece
Fungi	Dry rot, wet rot, blue stain, sap stain, brown rot, white rot, heart rot	**Dry rot** — fungus turns unventilated damp timber into dry powder. **Wet rot** — chemical decay from alternate wetting and drying
Insects	Beetles, termites (white ants), marine borers	Beetles leave powder-filled pin holes; termites eat the inside leaving a thin shell

Seasoning

- **Objects:** reduce weight, shrinkage and warping; increase strength, stiffness and durability; make the timber fit for paint, glue and preservatives; make it safe from fungi.
- **Natural seasoning** — **air seasoning**: timber stacked under a shed on a raised platform with gaps for air; cheap and gentle but slow (months), and moisture cannot be brought much below about 15–18 %. **Water seasoning**: logs kept in flowing water for 2–4 weeks to wash out sap, then air dried.
- **Artificial seasoning** — **kiln seasoning** (temperature, humidity and air flow controlled; days instead of months; any moisture content), boiling, **chemical (salt) seasoning**, electrical seasoning.
- Seasoned timber for general work has about **10 to 12 %** moisture, close to the equilibrium moisture content of the place where it will be used.

Preservative / process	What it is	Remember
Coal tar	Applied hot; cheap; bad smell, cannot be painted	Door and window frames built into walls
Creosote oil	Distilled from tar; applied under pressure	Sleepers, piles, poles. Doubles the life of timber
ASCU	Powder of hydrated arsenic pentoxide, copper sulphate and potassium dichromate in the ratio **1 : 3 : 4**, dissolved in water	Odourless, can be painted, protects against white ants
Chemical salts	Copper sulphate, zinc chloride, mercury chloride, sodium fluoride	Water-borne; can be painted over
Solignum paint	Applied hot with a brush	Against white ants
Charring	Lower ends of posts burnt to about 15 mm depth and quenched	For posts to be buried in the ground
Bethell (full-cell) process	Vacuum first, then preservative under pressure	Cells stay full — maximum absorption
Lowry / Rueping (empty-cell)	Pressure, then a final vacuum draws out the surplus	Deep penetration with less preservative
Abel's process	Sodium silicate, then kaolin, then sodium silicate again	**Fire-proofing** of timber

Plywood and other wood products

- **Veneer** — thin sheet (0.4 to 6 mm) peeled from a rotating log (rotary cutting) or sliced.
- **Plywood** — an **odd number** of veneers (3, 5, 7 ...) glued under pressure with the **grain of each layer at right angles** to the next. The outer layers are face plies, the middle is the core.
- **Why it is better than plain wood:** nearly equal strength in both directions, very little shrinkage or swelling, does not split when nailed near the edge, available in large sheets, can be bent to shape.
- **Block board / batten board / lamin board** — core of wooden strips (up to 25 mm, up to 80 mm and up to 7 mm wide) between veneers.
- **Fibre board** — pressed wood or vegetable fibre; **particle (chip) board** — wood chips bonded with resin; **hardboard** — dense fibre board.

4. Everyday analogy

Seasoning is like drying red chillies or papad: dried slowly in shade they stay flat and whole; dried too fast in harsh sun the skin sets first and they crack and curl — that is case hardening and warping. Plywood is like a woven mat: single reeds bend easily one way, but woven at right angles they resist bending both ways.

5. Confusing areas and exam traps

- **Heart shake** is widest at the centre; **star shake** is widest at the circumference. The 2023 question gave the description, not the name.
- **Dry rot** needs dampness and no ventilation — the name refers to the dry powder it leaves, not to dry conditions.
- Plywood always has an **odd** number of plies so that both faces have the same grain direction and the sheet stays balanced.
- ASCU ratio is 1 : 3 : 4 — arsenic, copper, chromium, in that order.
- Full-cell = Bethell; empty-cell = Lowry and Rueping.

6. Connections

Seasoning defects are the visible result of unequal shrinkage (Topic 1). Formwork for concrete (Topic 9) is made of plywood because it does not warp. Fungal decay needs moisture, air and warmth — the same three that corrode steel.

Memory code: "Heart from the heart, Star from the skin, Cup along the ring"

Heart from the heart	Heart shake starts at the pith; widest at the centre
Star from the skin	Star shake starts at the bark; widest at the circumference
Cup along the ring	Cup shake follows an annual ring
A-Cu-Cr = 1-3-4	ASCU: Arsenic pentoxide 1, Copper sulphate 3, potassium diChromate 4

Self-recall (close the notes, answer aloud, then check)

1. What is the difference between a check and a split?
   A check separates the fibres but does not go through the piece; a split goes right through from one face to the other.
2. Give three advantages of kiln seasoning over air seasoning.
   Much faster; moisture content can be brought to any required value; drying is controlled, so fewer defects and fungi and insects are killed.
3. Why does plywood have an odd number of plies?
   So that the two outer plies have the same grain direction and the sheet is balanced about its middle; it does not warp.

Quick check: 5 questions

1. A log shows cracks that start at the pith and run towards the sapwood, being widest at the centre and narrowing outwards. The defect is

   1. star shake
   2. cup shake
   3. heart shake
   4. rind gall

   1. Star shake is widest at the circumference.
   2. Cup shake is a curved crack along an annual ring.
   3. Correct. Heart shakes radiate from the pith and are widest there; they are caused by shrinkage of the interior of an over-mature tree.
   4. Rind gall is a swelling where a branch was cut.

2. Match List-I with List-II:

List-I	List-II
A. Bethell's process	1. Arsenic, copper and chromium salts in water
B. Abel's process	2. Full-cell pressure treatment
C. ASCU	3. Lower ends of posts buried in the ground
D. Charring	4. Fire-proofing of timber

   1. A-4, B-2, C-1, D-3
   2. A-2, B-4, C-1, D-3
   3. A-2, B-4, C-3, D-1
   4. A-1, B-4, C-2, D-3

   1. Bethell's is the full-cell pressure process, not fire-proofing.
   2. Correct. Bethell → full-cell; Abel → fire-proofing; ASCU → water-soluble salt preservative; charring → posts in the ground.
   3. ASCU is a preservative salt; charring is for buried post ends.
   4. Bethell's process is not a salt preservative.

3. **Assertion (A):** In plywood, the grain of each veneer is kept at right angles to the grain of the next.
Reason (R): Timber is much stronger and shrinks much less along the grain than across it.

1. Both A and R are true, and R is the correct explanation of A
2. Both A and R are true, but R is NOT the correct explanation of A
3. A is true, but R is false
4. A is false, but R is true

1. Correct. Crossing the layers means each direction always has some layers working along their grain, so strength is nearly equal and movement is restrained.
2. R is the direct reason for the cross-banding.
3. R is true.
4. A is true.

4. Consider the following statements:

I. Dry rot is caused by fungus in damp, poorly ventilated places.
II. Case hardening is a seasoning defect.
III. Wane is a defect caused by fungi.
IV. Air seasoning is slower than kiln seasoning.

Which of the above statements are correct?

1. I and II only
2. II, III and IV only
3. I, II, III and IV
4. I, II and IV only

1. Leaves out IV, which is true.
2. III is false.
3. Includes the false statement III.
4. Correct. Wane (III) is a conversion defect — the original rounded surface left on a sawn piece.

5. The preservative most suitable for railway sleepers and transmission poles, applied under pressure, is

1. solignum paint
2. sodium silicate
3. creosote oil
4. coal tar applied cold

1. A brush-applied paint against white ants.
2. Used in fire-proofing (Abel's process).
3. Correct. Creosote, forced in by a pressure process, is the standard preservative for sleepers, piles and poles.
4. Tar is applied hot, on embedded frame ends.

Topic 3: Bricks — Earth, Manufacture, Defects, IS Classification and Tests

Syllabus line: *Bricks: types, Indian standard classification, absorption, saturation factor.*

Asked in APPSC AEE 2023, Paper III Civil, 22 Aug Shift 2 (verified with official key): 2 questions

- Match the proportions of brick ingredients. The official key took: lime **less than 7 %**, ferric oxide **10 %**, silica **50–60 %**, magnesia **less than 1 %** (Q10). Textbooks (Rangwala) give lime up to about 5 % and iron oxide 5–6 %; in a matching question, fix silica and magnesia first.
- The deformation of the shape of bricks caused by **rain water falling on hot bricks** is called **chuffs** (Q12).

1. Basic idea

A **brick** is a block of clay that has been moulded, dried and burnt at about 900–1100 °C so that the clay particles fuse and become hard and permanent. Its small, uniform size lets one hand lift it, which is why brick masonry is the most common walling.

2. Why the ingredients matter

Good brick earth is a balance: **alumina** makes the clay plastic so that it can be moulded, and **silica** stops it from shrinking and cracking as it dries. Too much of either spoils the brick.

3. How it works

Ingredient	Proportion	Job	If in excess
Silica	50–60 %	Prevents cracking, shrinking and warping; gives uniform shape and durability	Brick becomes brittle and weak
Alumina	20–30 %	Gives plasticity for moulding	Brick shrinks, warps and cracks on drying
Lime	Small (up to about 5 %; less than 7 % in the 2023 key)	Reduces shrinkage; acts as a flux so that silica fuses	Brick melts and loses shape; lumps cause lime bursting
Oxide of iron	About 5–6 % (10 % in the 2023 key)	Gives the red colour; helps fusion; adds strength	Dark blue or black brick
Magnesia	Less than 1 %	Gives a yellow tint; reduces shrinkage	Decay of the brick

Harmful ingredients: lumps of lime, iron pyrites (decompose and split the brick), alkalis (cause **efflorescence** — white salt deposits), pebbles and stones, and organic matter (leaves the brick porous).

Manufacture in four stages

- **1. Preparation of clay:** unsoiling (remove top 20 cm) → digging → cleaning → **weathering** (exposed for weeks) → blending → **tempering** (kneading with water, in a **pug mill**).
- **2. Moulding:** hand moulding (ground-moulded or table-moulded) or machine moulding (plastic clay or dry clay). The mould is made **8–12 % larger** to allow for shrinkage. The **frog** is the 10–20 mm deep depression on the top face: it forms a key for mortar and carries the maker's mark.
- **3. Drying:** to about 2 % moisture, so that the bricks do not crack in the kiln.
- **4. Burning:** in a **clamp** (temporary, cheap, poor control) or a **kiln** — intermittent kilns, or continuous kilns: **Bull's trench kiln** (most common in India), **Hoffmann's kiln** (roofed, works in all seasons) and the tunnel kiln.
- **Standard (modular) size:** 190 × 90 × 90 mm; with mortar 200 × 100 × 100 mm, so **500 bricks per m³** of masonry. Traditional size is about 230 × 114 × 76 mm.

Defect	Cause
Chuffs	Rain water falling on hot bricks — the shape is deformed (2023 Q12)
Bloating	Spongy swollen mass on the surface — excess carbonaceous matter and sulphur
Black core	Bituminous matter or carbon not fully burnt out — brick heated too fast
Efflorescence	Soluble alkali salts come to the surface as a white deposit when the brick dries
Lime blowing	Lumps of lime slake later, swell and burst the brick
Over-burning / under-burning	Over-burnt (jhama) bricks are fused and misshapen; under-burnt (pilla) bricks are soft and yellowish
Checks, cracks, laminations	Excess water, drying too fast, or air trapped in the clay

Traditional class	Shape and burning	Water absorption	Crushing strength	Use
First class	Table-moulded, sharp edges, uniform colour, clear ringing sound	Not more than 20 %	At least 10.5 N/mm²	Exposed face work, load-bearing walls
Second class	Ground-moulded, slightly irregular	Not more than 22 %	At least 7 N/mm²	Walls that will be plastered
Third class	Ground-moulded, burnt in clamps, soft	Not more than 25 %	At least 3.5 N/mm²	Temporary and unimportant work
Over-burnt (jhama)	Fused, dark, distorted	—	—	Aggregate for foundations, floors, road metal

Indian Standard classification and tests (IS 1077, IS 3495)

- **IS 1077** classifies common burnt clay bricks by **average compressive strength**: classes 3.5, 5, 7.5, 10, 12.5, 15, 17.5, 20, 25, 30 and 35 (the number is the strength in N/mm²).
- **Water absorption** after 24 hours in cold water: not more than **20 %** for classes up to 12.5, and not more than **15 %** for higher classes. Absorption = (wet − dry)/dry × 100.

- **Efflorescence:** rated nil, slight (up to 10 % of the area), moderate (up to 50 %), heavy and serious. Not more than **moderate** up to class 12.5 and **slight** above.
- **Compressive strength test:** frog filled flush with 1 : 3 cement mortar, brick soaked, load applied on the flat face at 14 N/mm² per minute.
- **Dimension tolerance:** 20 bricks laid end to end must measure 3800 ± 80 mm in length and 1800 ± 40 mm in width and height.
- **Saturation coefficient** (C/B ratio) = absorption after 24 hours in cold water ÷ absorption after 5 hours of boiling. It shows how much of the pore space fills easily. The **lower** it is, the more empty space is left for freezing water to expand, so the brick is **more durable**; values lie between about 0.4 and 0.95, and below about 0.8 is good.
- **Field tests:** two bricks struck together give a clear metallic ring; a finger nail leaves no scratch; a brick dropped flat from 1 m does not break; a fractured face is uniform, with no lumps or holes.

4. Everyday analogy

Making bricks is like making chapati dough into something permanent. Too much fine flour (alumina) and the dough cracks as it dries; a little coarse flour (silica) holds the shape. Baking fixes it for ever — and sprinkling water on a hot tava makes it spit and distort, which is what rain does to hot bricks (chuffs).

5. Confusing areas and exam traps

- **Alumina → plasticity; silica → prevents shrinkage.** Questions swap these.
- **Chuffs** (rain on hot bricks) and **bloating** (excess carbon and sulphur) are both shape defects with different causes.
- IS 1077 classes are by **strength**; "first, second, third class" is the traditional classification by quality.
- A **lower** saturation coefficient is better. Do not read it like a strength.
- Efflorescence is caused by **alkali salts**, not by lime.

6. Connections

Brick strength and mortar strength together decide masonry strength (Topic 4). Water absorption is the reason bricks are soaked before laying: a dry brick sucks water out of the mortar (Topic 7). Fly-ash bricks use the pozzolanic reaction explained under cement (Topic 5).

Memory code: “Silica Stops Shrinking, Alumina Allows moulding”

Silica Stops Shrinking	Silica (50–60 %) prevents shrinkage, cracking and warping
Alumina Allows moulding	Alumina (20–30 %) gives plasticity
Chuffs = shower	Rain (a shower) on hot bricks deforms them
20 then 15	Water absorption: 20 % up to class 12.5, 15 % for higher classes

Self-recall (close the notes, answer aloud, then check)

1. What is the standard modular size of a brick and how many are needed per cubic metre of masonry?
   190 × 90 × 90 mm (200 × 100 × 100 mm with mortar); 500 bricks per m³.
2. Define the saturation coefficient of a brick and say whether a high or low value is better.
   Ratio of 24-hour cold-water absorption to 5-hour boiling-water absorption; a lower value means better durability (frost resistance).
3. What causes efflorescence and how is it rated?
   Soluble alkali salts carried to the surface by water; rated nil, slight, moderate, heavy, serious.

Quick check: 5 questions

1. A brick weighs 3.00 kg when dry and 3.45 kg after 24 hours of immersion in cold water. The same brick absorbs 20 % after 5 hours of boiling. Its water absorption and saturation coefficient are

   1. 13 % and 0.65
   2. 15 % and 0.75
   3. 15 % and 1.33
   4. 45 % and 0.44

   1. 13 % divides by the wet weight.
   2. Correct. Absorption = 0.45/3.00 = 15 %. Saturation coefficient = 15/20 = 0.75.
   3. The ratio is cold ÷ boiling, not the reverse.
   4. 0.45 kg is the mass of water, not a percentage.

2. Match List-I (ingredient of brick earth) with List-II (its main function):

List-I	List-II
A. Alumina	1. Prevents shrinkage, cracking and warping
B. Silica	2. Gives plasticity for moulding
C. Lime	3. Gives the red colour
D. Oxide of iron	4. Flux that helps silica to fuse

1. A-2, B-1, C-4, D-3
2. A-1, B-2, C-4, D-3
3. A-2, B-1, C-3, D-4
4. A-4, B-1, C-2, D-3

1. Correct. Alumina → plasticity; silica → prevents shrinkage and cracking; lime → flux; iron oxide → red colour.
2. Alumina and silica are swapped.
3. Lime is the flux; iron oxide gives the colour.
4. Alumina does not act as a flux.

3. **S1:** IS 1077 classifies common burnt clay bricks on the basis of their average compressive strength.
S2: For bricks of class 20, water absorption after 24 hours of cold immersion may be up to 20 %.

1. Both S1 and S2 are true
2. S1 is true and S2 is false
3. S1 is false and S2 is true
4. Both S1 and S2 are false

1. S2 is false.
2. Correct. S1 is true. For classes above 12.5 the limit is 15 %, so S2 is false.
3. S1 is true.
4. S1 is true.

4. Bloating of bricks, a spongy swollen mass on the surface, is caused by

1. rain water falling on hot bricks
2. lumps of lime in the clay
3. soluble alkali salts in the clay
4. excess carbonaceous matter and sulphur in the clay

1. That causes chuffs.
2. That causes lime blowing.
3. That causes efflorescence.
4. Correct. Gases from carbon and sulphur swell the surface.

5. Consider the following statements about a good burnt clay brick:
I. It gives a clear metallic ring when struck with another brick.
II. A finger nail should leave a clear scratch on its surface.
III. It should not break when dropped flat from a height of about 1 m.
IV. Its fractured surface should be uniform and free of lumps.
Which of the above statements are correct?

1. I and III only
2. II, III and IV only
3. I, III and IV only
4. I, II, III and IV

1. Leaves out IV, which is true.
2. II is false.
3. Correct. II is false: a well-burnt brick is too hard to be scratched by a finger nail.
4. Includes the false statement II.

Topic 4: Strength of Brick Masonry, Influence of Mortar — and Building Stones

Syllabus line: *Bricks: strength in masonry, influence of mortar strength on masonry strength.* Building stones are not named in the syllabus but were asked in 2023, so they are added here.

Asked in APPSC AEE 2023, Paper III Civil, 22 Aug Shift 2 (verified with official key): 1 question

- Match the property of stone with its test: hardness → **Mohs scale**; toughness → **resistance to hammering**; weather resistance → **acid test**; frost resistance → **Brard's test** (Q13).

1. Basic idea

Masonry is an assembly of units (bricks or stones) bonded with mortar. It is a composite, and its strength is **not** the strength of the brick. A wall of 10 N/mm² bricks may safely carry only a fraction of that.

2. Why masonry is weaker than its bricks

Mortar is softer than brick. Under vertical load the mortar in a joint tries to squeeze out sideways. Friction and bond make it drag the brick outwards with it, so the brick is put into **sideways tension** while it is being compressed. Brick is weak in tension, so the wall fails by **vertical splitting of the bricks**, long before the bricks would crush.

3. How it works

Factor	Effect on masonry strength
Strength of the brick	The main factor. Masonry strength rises with brick strength, but less than in proportion
Strength of the mortar	Stronger mortar gives stronger masonry, but the gain is **small**: roughly, doubling mortar strength raises masonry strength by only about a quarter. Mortar much stronger than the brick is wasteful and brittle, and cracks pass through the bricks
Thickness of joints	Thicker joints mean more squeezing and weaker masonry. Standard joint is about 10 mm
Bond and workmanship	Unfilled joints, bricks not laid level and continuous vertical joints reduce strength sharply
Slenderness ratio	Effective height (or length) ÷ effective thickness. Tall thin walls buckle; IS 1905 limits it to **27** for load-bearing walls in cement mortar
Eccentricity of load	Load off the centre adds bending and reduces the permissible stress
Wetting of bricks	Dry bricks suck water from the mortar and weaken the bond; bricks are soaked before laying

Rules to remember (IS 1905 — structural use of unreinforced masonry)

- The **basic compressive stress** of masonry is read from a table against two things: the **crushing strength of the unit** and the **type of mortar** (H1, H2, M1, M2, M3, L1, L2).
- Typical cement–sand mortars: **H1** 1 : 3 (about 10 N/mm²), **H2** 1 : 4 (7.5), **M1** 1 : 5 (5), **M2** 1 : 6 (3), **M3** 1 : 7 (1.5).
- The basic stress is then multiplied by reduction factors for **slenderness** and **eccentricity**, and by area and shape factors.
- Masonry strength is commonly only about **25 to 50 %** of the brick strength.
- A useful design rule: mortar should be **no stronger than needed**. A slightly weaker mortar accommodates small movements and confines cracking to the joints, which are easy to repair.
- **Bonds:** English bond (alternate courses of headers and stretchers — strongest); Flemish bond (headers and stretchers alternate in each course — better appearance). A **queen closer** is a brick cut lengthwise in half, placed next to the quoin header to break the vertical joints.

Building stones (asked in 2023)

Classification	Types	Examples
Geological	**Igneous** (cooled magma), **sedimentary** (deposited in layers), **metamorphic** (changed by heat and pressure)	Igneous: granite, basalt, trap. Sedimentary: sandstone, limestone, laterite. Metamorphic: marble, slate, quartzite, gneiss
Physical	Stratified, unstratified, foliated	Stratified: sandstone, limestone. Unstratified: granite. Foliated: slate
Chemical	Siliceous, argillaceous (clay), calcareous (lime)	Siliceous: granite, quartzite. Argillaceous: slate, laterite. Calcareous: limestone, marble

Metamorphic pairs: limestone → **marble**; shale → **slate**; sandstone → **quartzite**; granite → **gneiss**.

Property	Test	What is done
Hardness	**Mohs scale** (also Dorry's test)	Scratching: talc 1 ... diamond 10
Toughness	**Impact test** (resistance to hammering)	Number of blows a specimen withstands
Weather resistance	**Acid test**	Kept in dilute acid for a week; sharp edges and no deposit mean good weathering quality
Frost resistance	**Brard's test** (crystallisation test)	Soaked in sodium sulphate solution and dried repeatedly; loss of weight is found
Soluble or earthy matter	**Smith's test**	Stone chips shaken in clear water; muddiness shows earthy matter
Resistance to wear	**Attrition test** (Deval's machine)	Pieces rotated in a cylinder; percentage wear
Strength	Crushing test	40 mm cubes crushed; a good stone exceeds 100 N/mm²
Porosity	Water absorption test	Should not exceed about 0.6 % by weight for a good stone

4. Everyday analogy

Stack biscuits with a layer of jam between each pair and press down. The jam squeezes out and pulls the biscuits apart sideways; they crack long before they would be crushed on their own. A thicker layer of jam makes it worse. That is a brick wall with soft, thick mortar joints.

5. Confusing areas and exam traps

- Masonry strength is **lower** than both the brick strength and, for strong mortars, sometimes the mortar cube strength.
- Mortar strength has only a **minor** influence compared with brick strength — do not assume "richer mortar, proportionally stronger wall".
- Brard's test → frost; acid test → weathering; Smith's test → soluble matter. These three are swapped in matching questions.
- Marble is metamorphic and calcareous; granite is igneous and siliceous.

6. Connections

Design of brick masonry appears again in the RCC module ("design of brick masonry"). Mortar types are detailed in Topic 7. The splitting failure caused by sideways tension is the same mechanism used in the split-tensile test of concrete (Topic 9).

Memory code: “M-I-A-B” for stones and “soft joints split bricks”

M — Mohs	Hardness
I — Impact	Toughness (hammering)
A — Acid	Weathering
B — Brard	Frost
Soft joints split bricks	Mortar squeezes sideways and pulls the brick apart in tension

Self-recall (close the notes, answer aloud, then check)

1. Why does a brick wall fail by vertical splitting?
   The softer mortar expands sideways under load and drags the bricks outward, putting them in lateral tension; bricks are weak in tension.
2. What two things decide the basic compressive stress of masonry in IS 1905?
   The crushing strength of the masonry unit and the type (grade) of mortar.
3. From which rocks do marble, slate and quartzite form?
   Marble from limestone, slate from shale, quartzite from sandstone.

Quick check: 5 questions

1. **Assertion (A):** The compressive strength of brick masonry is much less than the compressive strength of the bricks used.
Reason (R): Mortar in the bed joints, being softer, tends to expand laterally and sets up tensile stresses in the bricks.

1. Both A and R are true, and R is the correct explanation of A
2. Both A and R are true, but R is NOT the correct explanation of A
3. A is true, but R is false
4. A is false, but R is true

1. Correct. The lateral tension splits the bricks vertically before they can crush.
2. R is exactly the mechanism behind A.
3. R is true.
4. A is true.

2. **Statement:** A wall is rebuilt with the same bricks, but the mortar is changed from 1 : 6 to 1 : 3 cement–sand.
Conclusion 1: The strength of the masonry increases.
Conclusion 2: The strength of the masonry increases in the same proportion as the strength of the mortar.

1. Only conclusion 1 is correct
2. Only conclusion 2 is correct
3. Both conclusions 1 and 2 are correct
4. Neither conclusion 1 nor 2 is correct

1. Correct. Masonry strength rises with mortar strength, but only slightly — far less than in proportion.
2. Conclusion 2 is false.
3. Conclusion 2 is false.
4. Conclusion 1 is true.

3. Match List-I (property of stone) with List-II (test):

List-I	List-II
A. Hardness	1. Brard's test
B. Frost resistance	2. Acid test
C. Weather resistance	3. Smith's test
D. Presence of soluble matter	4. Mohs scale

1. A-2, B-1, C-4, D-3
2. A-1, B-4, C-2, D-3
3. A-4, B-1, C-3, D-2
4. A-4, B-1, C-2, D-3

1. The acid test is not for hardness.
2. Hardness is tested on the Mohs scale.
3. Weathering uses the acid test; Smith's test is for soluble matter.
4. Correct. Hardness → Mohs; frost → Brard; weathering → acid; soluble matter → Smith.

4. Which of the following is a metamorphic rock formed from limestone?

1. Slate
2. Quartzite
3. Marble
4. Gneiss

1. Slate forms from shale.
2. Quartzite forms from sandstone.
3. Correct. Limestone changes to marble under heat and pressure.
4. Gneiss forms from granite.

5. Consider the following statements about brick masonry:

I. Increasing the thickness of mortar joints increases the strength of the masonry.

II. Bricks are soaked in water before laying so that they do not absorb water from the mortar.

III. The permissible stress reduces as the slenderness ratio of the wall increases.

IV. English bond has alternate courses of headers and stretchers.

Which of the above statements are correct?

1. I, II and III only
2. II, III and IV only
3. II and III only
4. I, II, III and IV

1. Includes the false statement I.
2. Correct. I is false: thicker joints weaken masonry. II, III and IV are true.
3. Leaves out IV, which is true.
4. Includes the false statement I.

Topic 5: Cement — Compounds, Hydration and Types

Syllabus line: *Cement: compounds of different types.*

Asked in APPSC AEE 2023, Paper III Civil, 22 Aug Shift 2 (verified with official key): 0 questions in this shift

- No direct question on cement chemistry came in this paper. The compounds and types below are among the most repeated facts in civil engineering exams, so study them fully.

1. Basic idea

Cement is a fine grey powder that reacts chemically with water and hardens into a stone-like mass. The reaction is called **hydration**. Cement does not "dry" hard — it needs water to harden, and it hardens even under water, so it is a **hydraulic** binder. Ordinary Portland cement (OPC) was patented by Joseph Aspdin in 1824.

2. Why the compounds matter

Cement is not one chemical. It is a mixture of four compounds, each with its own speed, strength and heat. By changing their proportions the manufacturer makes rapid-hardening, low-heat or sulphate-resisting cement. If you know what each compound does, you can work out every "type of cement" question.

3. How it works

Manufacture in one line

- Limestone (lime) + clay or shale (silica, alumina, iron oxide) → ground and blended (wet or dry process; the **dry process** is used today because it saves fuel) → burnt in a **rotary kiln at about 1400–1500 °C** → **clinker** → cooled and ground with **2–5 % gypsum** → cement.
- **Gypsum** is added to slow down the setting. Without it the cement would set at once (flash set).
- Oxide composition of OPC: **lime (CaO) 60–65 %**, silica 17–25 %, alumina 3–8 %, iron oxide 0.5–6 %, magnesia up to about 4 %, sulphur trioxide 1–3 %, alkalis under 1 %.
- Excess lime makes cement **unsound** (it expands later); too little lime reduces strength and makes it set quickly.

Bogue compound	Short form	Share	What it does	Heat
Tricalcium silicate	**C_3S**	about 40–55 %	Hydrates fast; gives **early strength** (first 7 days)	High (about 500 J/g)
Dicalcium silicate	**C_2S**	about 20–30 %	Hydrates slowly; gives **later strength** (after 7 days, up to a year)	Low (about 260 J/g)
Tricalcium aluminate	**C_3A**	about 8–12 %	Reacts first and fastest; causes **flash set** unless gypsum is present; little strength; attacked by **sulphates**	Highest (about 865 J/g)
Tetracalcium alumino-ferrite	**C_4AF**	about 6–10 %	Little strength; gives the grey colour; acts as a flux in the kiln	Moderate (about 420 J/g)

Hydration

- C_3S and C_2S + water → **calcium silicate hydrate (C–S–H gel)** + **calcium hydroxide $Ca(OH)_2$**. The gel is the glue that gives strength.
- $Ca(OH)_2$ keeps concrete highly alkaline (pH about 12.5–13), which **protects the steel from rusting**; but it is the part attacked by sulphates and acids and leached by water. C_2S produces less of it than C_3S.
- **Water needed for complete hydration ≈ 38 % of the weight of cement**: about 23 % combines chemically (bound water) and about 15 % is held in the gel pores (gel water).
- Hydration gives out heat — the **heat of hydration**. In a thick dam the inside heats up, cools later and cracks, so low-heat cement is used there.
- **Pozzolana** (fly ash, surkhi, silica fume) has no cementing value alone, but reacts with $Ca(OH)_2$ and water to form more C–S–H gel.

Type of cement	How it is made different	Where it is used
OPC grades 33, 43, 53 (IS 269)	Grade = 28-day strength of the standard mortar cube in N/mm²	General construction; 53 grade for high-strength and prestressed work
Rapid hardening (IS 8041)	More C_3S, ground finer. 3-day strength equals the 7-day strength of OPC	Road repairs, early removal of formwork, cold weather
Low heat (IS 12600)	Less C_3S and C_3A, more C_2S. Gains strength slowly	Mass concrete — dams, thick foundations
Sulphate resisting (IS 12330)	C_3A kept below 5 %	Foundations in sulphate soils, sewage works, marine work
Portland pozzolana (PPC) (IS 1489)	OPC clinker + 15–35 % fly ash or calcined clay	General and mass concrete, marine and hydraulic structures; lower heat, better durability, slower early strength
Portland slag (PSC) (IS 455)	Clinker + 25–70 % granulated blast-furnace slag	Marine works, mass concrete
Quick setting	Less gypsum, a little aluminium sulphate added. Sets in minutes	Concreting under water or in running water
High alumina (IS 6452)	Made from bauxite and lime; very high early strength and heat	Refractory and chemical works; not for mass concrete
Hydrophobic (IS 8043)	Ground with a water-repelling film (oleic or stearic acid)	Long storage in humid places
White (IS 8042)	Iron oxide below 1 %; fired with oil	Decorative work, tile joints

Setting and hardening are different. Setting is the change from a plastic paste to a solid (stiffening). **Hardening** is the gain of strength after setting. **False set** is early stiffening without much heat; remixing without adding water restores plasticity. **Flash set** is immediate stiffening with heat, caused by C_3A when gypsum is insufficient; it cannot be undone.

4. Everyday analogy

Think of a relay team. C_3A is the over-excited starter who would finish the race in a second and spoil it, so gypsum holds his hand. C_3S is the sprinter who gives the early lead (early strength). C_2S is the marathon runner who keeps adding distance for a year (later strength). C_4AF mostly carries the team colours (grey).

5. Confusing areas and exam traps

- Early strength → **C_3S**; later strength → **C_2S**; flash set and most heat → **C_3A**.
- Low-heat cement has **less** C_3S and C_3A; rapid-hardening has **more** C_3S. Sulphate-resisting has **less C_3A**.
- Gypsum **retards** setting; it does not add strength.
- Rapid-hardening cement is not quick-setting cement: its setting times are the same as OPC; only the strength comes sooner.
- Water for full hydration is 38 % (23 % bound + 15 % gel), not 23 % alone.

6. Connections

The 38 % figure explains why very low water–cement ratios leave unhydrated cement (Topic 8). $Ca(OH)_2$ and alkalinity return in durability and carbonation (Topic 9). Blended cements (PPC, PSC) need longer curing (Topic 9). Low-heat cement links to dams in the Irrigation module.

Memory code: “3S early, 2S late, 3A hot and hasty”

3S early	C_3S — early strength
2S late	C_2S — later strength, low heat
3A hot and hasty	C_3A — highest heat, flash set, sulphate attack
38 = 23 + 15	Water for full hydration: 23 % bound + 15 % gel water

Self-recall (close the notes, answer aloud, then check)

1. Why is gypsum added to clinker?
   To retard the setting of cement by controlling the very fast reaction of C_3A (prevents flash set).
2. Which compounds are reduced in low-heat cement, and which is increased?
   C_3S and C_3A are reduced; C_2S is increased.
3. What are the two products of hydration of the calcium silicates?
   Calcium silicate hydrate (C–S–H gel), which gives strength, and calcium hydroxide.

Quick check: 5 questions

1. Match List-I (compound) with List-II (role):

List-I	List-II
A. C_3S	1. Later strength, low heat
B. C_2S	2. Grey colour, little strength
C. C_3A	3. Early strength
D. C_4AF	4. Flash set and highest heat of hydration

1. A-4, B-1, C-3, D-2
2. A-1, B-3, C-4, D-2
3. A-3, B-1, C-2, D-4
4. A-3, B-1, C-4, D-2

1. C_3S does not cause flash set.
2. C_3S and C_2S are swapped.
3. C_3A, not C_4AF, causes flash set.
4. Correct. C_3S → early strength; C_2S → later strength; C_3A → flash set and most heat; C_4AF → grey colour, little strength.

2. **Statement:** A cement is to be made for the mass concrete of a gravity dam.
Conclusion 1: Its C_3A and C_3S contents should be kept low.
Conclusion 2: It will gain strength faster than ordinary Portland cement.

1. Only conclusion 1 is correct
2. Only conclusion 2 is correct
3. Both conclusions 1 and 2 are correct
4. Neither conclusion 1 nor 2 is correct

1. Correct. Low heat needs less C_3A and C_3S; with more C_2S the strength comes more slowly, so conclusion 2 is false.
2. Conclusion 2 is false.
3. Conclusion 2 is false.
4. Conclusion 1 is true.

3. The approximate quantity of water required for the complete hydration of 50 kg of cement is

1. 11.5 litres
2. 15 litres
3. 19 litres
4. 25 litres

1. This is only the chemically bound water (23 %).
2. Too low.
3. Correct. 38 % of 50 kg = 19 kg = 19 litres (23 % bound + 15 % gel water).
4. That is a water–cement ratio of 0.5, more than needed for hydration.

4. **S1:** Sulphate-resisting cement has a low C_3A content.
S2: Rapid-hardening cement has shorter initial and final setting times than ordinary Portland cement.

1. Both S1 and S2 are true
2. S1 is true and S2 is false
3. S1 is false and S2 is true
4. Both S1 and S2 are false

1. S2 is false.
2. Correct. C_3A is the compound attacked by sulphates, so it is limited to 5 %. Rapid-hardening cement gains strength sooner but its setting times are the same as OPC.
3. S1 is true.
4. S1 is true.

5. Consider the following statements:
I. Gypsum is added to clinker to retard setting.
II. Calcium hydroxide produced in hydration keeps concrete alkaline and protects steel.
III. Pozzolana reacts with calcium hydroxide to form additional cementing gel.
IV. In low-heat cement the proportion of C_2S is reduced.

Which of the above statements are correct?

1. I, II, III and IV
2. II and III only
3. I, III and IV only
4. I, II and III only

1. Includes the false statement IV.
2. Leaves out I, which is true.
3. IV is false.
4. Correct. IV is false: C_2S is increased in low-heat cement, while C_3S and C_3A are reduced.

Topic 6: Cement — Setting Times, Soundness, Fineness and Strength Tests

Syllabus line: *Cement: setting times, strength.*

Asked in APPSC AEE 2023, Paper III Civil, 22 Aug Shift 2 (verified with official key): 0 questions in this shift

- No question on cement tests came in this paper. The limiting values below (30 minutes, 600 minutes, 10 mm, 10 %, the three strengths of each grade) are standard one-mark facts, and Topic 6 is the usual source of 'match the test with the apparatus' questions.

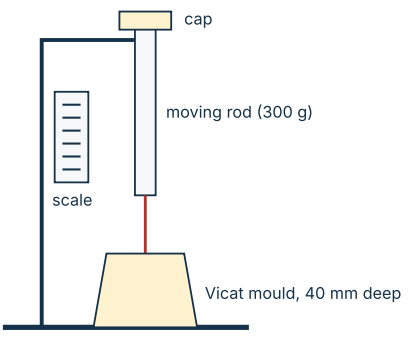
1. Basic idea

Before cement is used, a sample is tested to confirm five things: it is ground fine enough, it stays workable long enough, it hardens soon enough, it will not expand later, and it reaches its grade strength. The tests are given in **IS 4031** and the limits in the code of each cement (IS 269 for OPC).

2. Why each test exists

Each test answers a site question. **Initial setting time**: do I have enough time to mix, carry, place and compact? **Final setting time**: will it harden soon enough for the work to proceed? **Soundness**: will the hardened mass crack later by expanding? **Fineness**: will it hydrate fast enough? **Strength**: is it really 33, 43 or 53 grade?

3. How it works



Three attachments

Plunger, 10 mm diameter
Standard consistency: stops 5–7 mm from the bottom

Needle, 1 mm square
Initial set: stops 5 ± 0.5 mm from the bottom

Needle with 5 mm annular collar
Final set: needle marks the surface, collar does not

Water used: 0.85 P for setting-time tests, 0.78 P for the soundness test, where P is the standard consistency (usually 26–33 %).

One instrument, three attachments, three tests. P is found first, because every other test uses a paste made with a fixed fraction of P.

Test	Apparatus / method	Limit for OPC	What it controls
Fineness	Sieving on a **90 micron** sieve; or **Blaine's air permeability** apparatus (specific surface)	Residue not more than **10 %**; specific surface not less than **225 m²/kg** (rapid hardening 325)	Finer cement hydrates faster and gains strength sooner, but gives more heat and shrinkage
Standard (normal) consistency	Vicat apparatus with the 10 mm plunger	Plunger settles to 5–7 mm from the bottom; P is about 26–33 %	Fixes the water for the other tests
Initial setting time	Vicat, 1 mm square needle; paste with 0.85 P water	Not less than **30 minutes**	Time available for mixing, transporting and placing
Final setting time	Vicat, needle with annular collar	Not more than **600 minutes** (10 hours)	Paste has lost all plasticity; hardening begins
Soundness	**Le Chatelier** apparatus (split brass cylinder with two pointers; boiled for 3 hours) — detects free **lime**. **Autoclave** test — detects lime and **magnesia**	Expansion not more than **10 mm**; autoclave expansion not more than **0.8 %**	Unsoundness = late expansion from excess free lime, magnesia or sulphate
Compressive strength	Cubes of **70.6 mm** side (face area 50 cm²); mortar **1 : 3** with standard sand; water = **(P/4 + 3) %** of the weight of cement plus sand	See the next table	The grade of the cement
Tensile strength	Briquette test; water (P/5 + 2.5) %	Rarely specified now	—
Heat of hydration	Calorimeter	Low-heat cement: not more than 272 kJ/kg at 7 days and 314 kJ/kg at 28 days	Cracking in mass concrete
Specific gravity	Le Chatelier flask with **kerosene**	About **3.15**	Used in mix design

Minimum compressive strength (N/mm²)	3 days	7 days	28 days
OPC 33 grade	16	22	33
OPC 43 grade	23	33	43
OPC 53 grade	27	37	53
Rapid hardening	27 (and 16 at 1 day)	—	—
Low heat	10	16	35

Other values worth knowing

- Setting times of special cements: **low heat** — initial not less than 60 minutes; **quick setting** — initial 5 minutes, final 30 minutes; rapid hardening, PPC, slag — same as OPC (30 and 600 minutes).
- Test room: temperature 27 ± 2 °C, relative humidity 65 ± 5 %.
- **Bulk density** of cement = 1440 kg/m³, so one **50 kg bag = about 0.035 m³ (35 litres)**.
- **Storage:** cement absorbs moisture and loses strength — roughly 20 % in 3 months, 30 % in 6 months and 40 % in 12 months. Bags are stacked on a raised dry platform, 30 cm clear of walls, not more than 10 bags high.
- **Field tests:** uniform greenish-grey colour; feels cool when the hand is thrust in; smooth, not gritty, between the fingers; a handful thrown into water floats briefly before sinking; no hard lumps.

4. Everyday analogy

Think of setting curd. There is a period when you can still stir the milk and pour it into cups (before initial set). After a while it must not be disturbed (initial set has passed). Later it is fully set and can be carried (final set). Fineness is like powdered sugar against sugar crystals: the powder dissolves and acts much faster because it has more surface.

5. Confusing areas and exam traps

- Initial setting time is a **minimum** (not less than 30 minutes); final setting time is a **maximum** (not more than 600 minutes).
- **Le Chatelier** detects unsoundness due to free **lime only**; the **autoclave** test covers lime and magnesia.
- Water: **0.85 P** for setting time, **0.78 P** for soundness, **(P/4 + 3) %** for strength cubes.
- Cement cubes are **70.6 mm**; concrete cubes are **150 mm**.

- Le Chatelier has given his name to two things: the soundness apparatus and the specific-gravity flask.

6. Connections

Setting time decides how far ready-mixed concrete can travel and why retarders are added in summer (Topic 8). Strength of cement feeds directly into mix design (Topic 10). Loss of strength in storage is why site engineers ask for the week of manufacture printed on the bag.

Memory code: "30 in, 600 out, 10 and 10"

30 in	Initial setting time: at least 30 minutes
600 out	Final setting time: at most 600 minutes
10 and 10	Soundness expansion at most 10 mm; residue on the 90 micron sieve at most 10 %
85 – 78	0.85 P for setting time, 0.78 P for soundness

Self-recall (close the notes, answer aloud, then check)

1. What is standard consistency?
   The percentage of water at which the 10 mm Vicat plunger penetrates the paste to a point 5 to 7 mm from the bottom of the mould.
2. Which test detects unsoundness due to magnesia?
   The autoclave test (the Le Chatelier test detects free lime only).
3. Give the 3-, 7- and 28-day strengths required of 43 grade OPC.
   23, 33 and 43 N/mm².

Quick check: 5 questions

1. The standard consistency of a cement is 30 %. The quantity of water to be added to 400 g of the cement for the initial setting time test is

   1. 93.6 g
   2. 120 g
   3. 102 g
   4. 42 g

   1. This uses 0.78 P, which is for the soundness test.
   2. This is P itself, used for the consistency test.
   3. Correct. 0.85 × 30 % = 25.5 %; 25.5 % of 400 g = 102 g.
   4. This is (P/4 + 3) % of 400 g, the rule for strength cubes.

2. Match List-I (test on cement) with List-II (apparatus):

List-I	List-II
A. Soundness	1. Vicat apparatus with needle
B. Setting time	2. Flask with kerosene
C. Fineness	3. Le Chatelier split cylinder
D. Specific gravity	4. Blaine's air permeability apparatus

   1. A-1, B-3, C-4, D-2
   2. A-3, B-1, C-4, D-2
   3. A-3, B-1, C-2, D-4
   4. A-3, B-4, C-1, D-2

   1. Soundness is not found with the Vicat apparatus.
   2. Correct. Soundness → Le Chatelier apparatus; setting time → Vicat; fineness → Blaine; specific gravity → Le Chatelier flask with kerosene.
   3. Fineness uses Blaine's apparatus; the flask is for specific gravity.
   4. Setting time uses the Vicat apparatus.

3. **S1:** For ordinary Portland cement, the initial setting time should not be less than 30 minutes.
S2: The Le Chatelier test detects unsoundness caused by both free lime and magnesia.

1. Both S1 and S2 are true
2. S1 is true and S2 is false
3. S1 is false and S2 is true
4. Both S1 and S2 are false

1. S2 is false.
2. Correct. S1 is the code limit. The Le Chatelier test detects free lime only; magnesia needs the autoclave test.
3. S1 is true.
4. S1 is true.

4. The cubes used for finding the compressive strength of cement have a side of

1. 70.6 mm
2. 50 mm
3. 100 mm
4. 150 mm

1. Correct. A 70.6 mm cube has a face area of 50 cm².
2. Not the IS size for cement.
3. A size sometimes used for concrete with small aggregate.
4. The standard concrete cube.

5. **Assertion (A):** Finer cement gains strength faster at early ages.
Reason (R): Finer grinding gives a larger surface area for the reaction with water.

1. Both A and R are true, and R is the correct explanation of A
2. Both A and R are true, but R is NOT the correct explanation of A
3. A is true, but R is false
4. A is false, but R is true

1. Correct. Hydration starts at the particle surface, so more surface means faster hydration.
2. R is the direct reason.
3. R is true.
4. A is true.

Topic 7: Mortar — Ingredients, Proportions, Water Demand, Plastering; Lime

Syllabus line: *Cement mortar: ingredients, proportions, water demand, mortars for plastering and masonry.*

Asked in APPSC AEE 2023, Paper III Civil, 22 Aug Shift 2 (verified with official key): 0 questions in this shift

- No direct question came from mortar in this paper. Proportions for different works, bulking of sand and the quantity of cement in a given volume of mortar are the usual questions.

1. Basic idea

Mortar is a workable paste of a **binder** (cement or lime), **fine aggregate** (sand) and **water**. It beds and joins bricks or stones, and as plaster it covers them. Concrete is mortar plus coarse aggregate.

2. Why each ingredient is there

The **binder** glues. The **sand** gives bulk cheaply, and — more important — prevents the shrinkage cracks that a rich paste would develop. The **water** hydrates the cement and makes the mix workable. Sand takes no part in the chemistry; it is an inert filler.

3. How it works

Work	Usual cement : sand proportion (by volume)
Masonry in general (walls above plinth)	1 : 6
Load-bearing masonry, masonry in foundations and plinth	1 : 4 to 1 : 5
Masonry in very damp conditions, reinforced brickwork	1 : 3
Plastering — internal walls	1 : 5 to 1 : 6 (12 mm thick)
Plastering — external walls	1 : 4 (in two coats, 20 mm)
Plastering — ceilings and concrete surfaces	1 : 3 (6 mm)

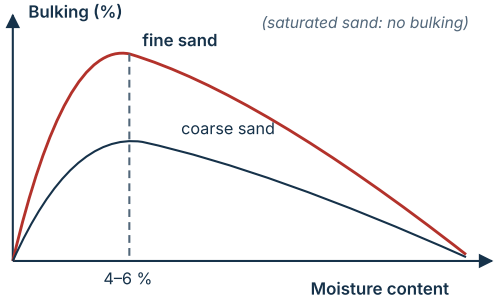
Work	Usual cement : sand proportion (by volume)
Pointing	1 : 1 to 1 : 3
Damp-proof course	1 : 2 with a water-proofing compound

Water demand and workability of mortar

- Mortar needs **more water than concrete** of the same richness: it must spread under the trowel, and part of its water is sucked out by the bricks.
- Water demand **rises** with finer sand, richer mix and higher temperature. Finer sand has more surface to be wetted.
- **Water retentivity** is the ability of mortar to hold its water against the suction of the brick. Poor retention leaves too little water for hydration and a weak bond. **Lime** improves workability and water retention — the reason for cement–lime (**gauged** or composite) mortar such as 1 : 1 : 6.
- Too much water lowers strength and causes shrinkage cracks; too little gives a harsh mix that does not bond.
- Cement mortar should be used **within 30 minutes** of adding water (before the initial set). Lime mortar can wait much longer.
- Strength of mortar is tested on cubes (IS 2250); it must be kept wet (**cured**) for at least 7 days.

Sand

- **Sources:** pit sand (sharp, angular — good for mortar), river sand (rounded, fine — good for plaster), sea sand (contains salts — causes efflorescence, avoided). Crushed-stone sand is now widely used.
- **Bulking of sand:** a film of water round each grain pushes the grains apart, so damp sand occupies more volume than dry sand. Bulking is greatest, about **20–40 %**, at **4–6 % moisture**. **Fine sand bulks more** than coarse sand. Fully saturated (inundated) sand returns to its dry volume.
- Bulking % = (bulked volume − dry volume) ÷ dry volume × 100. When sand is measured by volume, extra is taken to allow for bulking.
- **Fineness modulus** of sand: fine 2.2–2.6, medium 2.6–2.9, coarse 2.9–3.2.
- Silt content should not exceed about 8 % by volume (field test in a measuring cylinder); organic impurities are detected by the colour test with sodium hydroxide.



Dry volume rule

Dry materials shrink when mixed with water.

Dry volume of mortar ≈ 1.3 × wet volume

Dry volume of concrete ≈ 1.52–1.54 × wet volume

Example: 1 m³ of 1 : 6 mortar
cement = 1.3 ÷ 7 = 0.186 m³
= 0.186 × 1440 ≈ 267 kg ≈ 5.3 bags

Left: damp sand swells, most at 4–6 % moisture, and fine sand swells most. Right: the rule used in estimating materials for mortar.

Plastering and pointing

- **Objects of plastering:** a smooth, hygienic surface; protection from rain; a base for paint; hiding poor workmanship.
- **Coats:** three-coat work — **rendering coat** (first, 9–10 mm), **floating coat** (second, 6–9 mm), **finishing coat** (2–3 mm). Two-coat work omits the floating coat. Each coat is kept rough to key the next.
- **Thickness:** 12 mm on the fair (smooth) face of a brick wall, 15 mm on the rough face, 20 mm in two coats outside, 6 mm on concrete ceilings.
- **Defects: blistering** (small swellings), **crazing** (hair cracks), **cracking**, **efflorescence** (white salts), **flaking** and **peeling** (loss of bond), **popping** (conical holes caused by particles that expand, such as unslaked lime), **rust stains**.
- **Pointing** is finishing the joints only, raked out to 10–20 mm and filled with rich mortar. Types: flush, recessed, struck (weathered), tuck, V-pointing, beaded.

Lime (the older binder — still asked)

Term	Meaning
Calcination	Heating limestone: $CaCO_3$ → $CaO + CO_2$. The product CaO is **quick lime**
Slaking	Adding water to quick lime: $CaO + H_2O$ → $Ca(OH)_2$ + heat. The product is **slaked** or **hydrated lime**
Setting of fat lime	$Ca(OH)_2 + CO_2$ (from air) → $CaCO_3$. It sets by **carbonation**, slowly, and only in air
Fat (rich, white) lime	Nearly pure. Slakes vigorously, swells to 2–2.5 times, very plastic, sets slowly, does not set under water. For plaster and whitewash. IS 712 **Class C**
Hydraulic lime	Contains clay (5–30 %). Slakes slowly, **sets under water**, stronger. For masonry. IS 712 **Class A** (eminently hydraulic) and **Class B** (semi-hydraulic)
Poor (lean) lime	More than 30 % clay; slakes very slowly; poor mortar
Other IS 712 classes	D — magnesian (dolomitic) lime; E — kankar lime; F — siliceous dolomitic lime
Surkhi	Powdered burnt brick — a pozzolana added to lime mortar to give it hydraulic properties

4. Everyday analogy

Cement paste alone is like pure jaggery syrup — sticky, costly, and it cracks as it dries. Add puffed rice (sand) and you get a firm, cheap laddu that holds its shape. Bulking of sand is like damp hair: slightly wet, it stands up and takes more room; soaked, it lies flat again.

5. Confusing areas and exam traps

- Bulking is greatest at **4–6 %** moisture, and **fine** sand bulks more. Saturated sand does not bulk.
- Sand does **not** add strength to mortar; richer mortar (less sand) is stronger but shrinks and cracks more.
- Fat lime hardens by absorbing **CO_2 from air**; hydraulic lime and cement harden by reacting with **water**.
- Dry volume of mortar is about 1.3 times the wet volume; forgetting this factor gives a wrong cement quantity.
- Popping (conical holes) and blistering (swellings) are different plaster defects.

6. Connections

Mortar grade and masonry strength were linked in Topic 4. Bulking allowance is needed again when concrete is batched by volume (Topic 8). The pozzolanic action of surkhi is the same reaction as fly ash in PPC (Topic 5).

Memory code: "6 for walls, 4 outside, 3 above, 2 for damp"

6 for walls	1 : 6 — general masonry and internal plaster
4 outside	1 : 4 — external plaster and load-bearing masonry
3 above	1 : 3 — ceiling plaster
2 for damp	1 : 2 — damp-proof course
Bulk at 4–6	Sand bulks most at 4–6 % moisture; fine sand most

Self-recall (close the notes, answer aloud, then check)

1. Why is sand added to cement in mortar?
   To give bulk economically and to prevent shrinkage cracking; it is an inert filler.
2. What is bulking of sand and when is it greatest?
   Increase in volume of damp sand due to water films round the grains; greatest (20–40 %) at about 4–6 % moisture, more for fine sand.
3. How does fat lime set?
   By carbonation: calcium hydroxide absorbs carbon dioxide from the air and becomes calcium carbonate.

Quick check: 5 questions

1. The quantity of cement required for 2 m^3 of 1 : 4 cement mortar, taking the dry volume as 1.3 times the wet volume and the density of cement as 1440 kg/m^3, is about

1. 576 kg
2. 520 kg
3. 936 kg
4. 749 kg

1. This omits the factor 1.3.
2. This is the volume 0.52 m^3 read as kilograms.
3. This divides by 4 instead of 5 parts.
4. Correct. Dry volume = 2.6 m^3; cement = 2.6/5 = 0.52 m^3 = 0.52 × 1440 ≈ 749 kg (about 15 bags).

2. **Statement:** Damp sand containing about 5 % moisture is measured by volume for a mortar mix.
Conclusion 1: The actual quantity of sand in the measured volume is less than in the same volume of dry sand.
Conclusion 2: The mortar produced will be leaner than intended.

1. Only conclusion 1 is correct
2. Only conclusion 2 is correct
3. Both conclusions 1 and 2 are correct
4. Neither conclusion 1 nor 2 is correct

1. Correct. Bulked sand holds less sand per box, so conclusion 1 is true. Less sand per part of cement makes the mix richer, not leaner, so conclusion 2 is false.
2. Conclusion 1 is true and conclusion 2 is false.
3. Conclusion 2 is false: the mix becomes richer, not leaner.
4. Conclusion 1 is true.

3. Match List-I (work) with List-II (usual cement mortar):

List-I	List-II
A. General brick masonry	1. 1 : 2
B. External plastering	2. 1 : 3
C. Damp-proof course	3. 1 : 4
D. Ceiling plastering	4. 1 : 6

1. A-4, B-3, C-2, D-1
2. A-3, B-4, C-1, D-2
3. A-4, B-3, C-1, D-2
4. A-1, B-3, C-4, D-2

1. Damp-proof course is 1 : 2; ceiling plaster 1 : 3.
2. General masonry is 1 : 6, external plaster 1 : 4.
3. Correct. General masonry 1 : 6; external plaster 1 : 4; damp-proof course 1 : 2; ceiling plaster 1 : 3.
4. General masonry does not need 1 : 2.

4. **Assertion (A):** Lime is added to cement mortar to make gauged mortar for masonry.
Reason (R): Lime improves the workability and the water-retaining capacity of the mortar.

1. Both A and R are true, and R is the correct explanation of A
2. Both A and R are true, but R is NOT the correct explanation of A
3. A is true, but R is false
4. A is false, but R is true

1. Correct. Better water retention keeps water in the joint for hydration and gives a better bond.
2. R is the reason.
3. R is true.
4. A is true.

5. Consider the following statements:

I. Fat lime sets under water.

II. Hydraulic lime contains clay and sets under water.

III. Slaking of quick lime gives out heat.

IV. Popping in plaster is caused by particles that expand after the plaster has set.

Which of the above statements are correct?

1. I, II and III only
2. II and III only
3. I, II, III and IV
4. II, III and IV only

1. Includes the false statement I.
2. Leaves out IV, which is true.
3. Includes the false statement I.
4. Correct. I is false: fat lime sets only in air, by carbonation.

Topic 8: Concrete — Ingredients, Aggregates, Admixtures and the Water–Cement Ratio

Syllabus line: *Concrete: importance of w/c ratio, strength, ingredients including admixtures.*

Asked in APPSC AEE 2023, Paper III Civil, 22 Aug Shift 2 (verified with official key): 1 question

- Which aggregates are generally used for **high-strength concrete**? Key: **angular** aggregates — their rough faces and interlocking give a better bond with the cement paste (Q18).

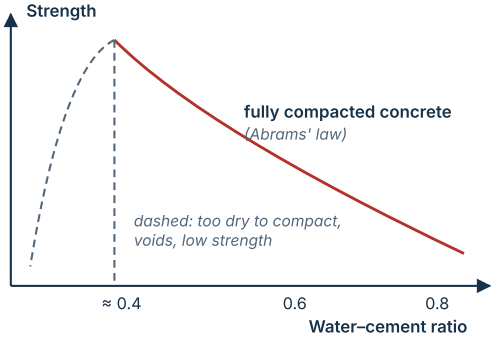
1. Basic idea

Concrete is cement, fine aggregate (sand), coarse aggregate (stone) and water mixed in fixed proportions, which hardens into an artificial stone. Aggregates fill about 70–75 % of the volume; cement paste coats every particle and fills the gaps. **M20** means a mix whose **characteristic compressive strength** is 20 N/mm², measured on 150 mm cubes at 28 days.

2. Why the water–cement ratio is the most important number

Cement needs water equal to only about 0.38 of its weight to hydrate fully. Any extra water is added just to make the mix workable. That extra water does not combine; it later dries out and leaves fine **capillary pores**. More pores means lower strength, more permeability and less durability. So for a fully compacted concrete, **strength depends mainly on the water–cement ratio** — the lower the ratio, the stronger the concrete. This is **Abrams' law** (Duff Abrams, 1918).

3. How it works



Read the graph

Right of the peak: more water, more pores, weaker concrete.

Left of the peak: the mix cannot be compacted, air voids remain, strength falls again.

5 % air voids cut strength by about 30 %; 10 % voids by about 60 %.

Abrams: $S = A \div B^{x}$, where x = w/c ratio.
Powers: $S = 240 \times (\text{gel/space ratio})^{3}$ N/mm².

Strength falls as the water–cement ratio rises, provided the concrete is fully compacted. Very dry mixes lose strength because they cannot be compacted.

Shape of aggregate	Voids	Effect
Rounded (river gravel)	Least, about 32–33 %	Best workability, least cement paste needed; poorer bond and interlock
Irregular (pit gravel)	About 35–37 %	In between
Angular (crushed rock)	Most, about 38–41 %	Best bond and interlock → preferred for **high-strength concrete** and pavements; needs more paste for the same workability

Shape of aggregate	Voids	Effect
Flaky	—	Least dimension (thickness) less than **0.6** times the mean dimension. Weak, harms workability
Elongated	—	Greatest dimension (length) more than **1.8** times the mean dimension

Aggregates — facts and tests (IS 383, IS 2386)

- **Fine aggregate** passes the **4.75 mm** sieve; **coarse aggregate** is retained on it. Sand is graded into **Zones I to IV** by the percentage passing the 600 micron sieve — Zone I is the coarsest, Zone IV the finest.
- **Fineness modulus** = sum of cumulative percentages retained on the standard sieves (80, 40, 20, 10, 4.75, 2.36, 1.18 mm, 600, 300, 150 micron) ÷ 100. A larger value means a coarser aggregate: sand about 2–3.5; coarse aggregate about 5.5–8.
- **Grading:** a well-graded aggregate has all sizes, leaves the fewest voids and needs the least paste.
- **Crushing value, impact value:** not more than **30 %** for wearing surfaces (roads, runways) and **45 %** for other concrete. **Los Angeles abrasion value:** not more than 30 % for wearing surfaces and 50 % otherwise.
- **Maximum size:** as large as possible (usually 20 mm for RCC), but not more than one-quarter of the least thickness of the member and at least 5 mm less than the clear spacing of bars and the cover.
- **Alkali–aggregate reaction:** reactive silica in some aggregates reacts with the alkalis of cement and forms a swelling gel that cracks the concrete. Controlled by low-alkali cement (alkali below 0.6 %), pozzolana and non-reactive aggregate.
- **Water** (IS 456): fit to drink in general; **pH not less than 6**; limits — organic solids 200 mg/l, inorganic 3000 mg/l, sulphates 400 mg/l, chlorides 2000 mg/l for plain concrete and **500 mg/l for reinforced concrete**, suspended matter 2000 mg/l. Sea water is not used for reinforced or prestressed concrete.

Admixture (IS 9103)	Action	Examples	Use
Plasticiser (water reducer)	Disperses cement grains; same workability with 5–15 % less water, or more workability with the same water	Lignosulphonates	General
Superplasticiser (high-range water reducer)	Reduces water by 15–30 %	Sulphonated naphthalene or melamine formaldehyde, polycarboxylate ether	High-strength, pumped and self-compacting concrete
Retarder	Delays setting	Gypsum, sugar (about 0.05 %), tartaric acid	Hot weather, long haul of ready-mixed concrete, large pours
Accelerator	Speeds setting and early strength	Calcium chloride (not for reinforced or prestressed concrete — it corrodes steel), triethanolamine	Cold weather, urgent repairs
Air-entraining agent	Forms millions of tiny separate air bubbles (3–6 %)	Vinsol resin, natural wood resins	Resistance to freezing and thawing; better workability; strength falls a little
Mineral admixtures	Pozzolanic or cementing powders that replace part of the cement	Fly ash, ground granulated blast-furnace slag, silica fume, metakaolin, rice-husk ash	Durability, lower heat; silica fume for very high strength

Grade	Nominal mix (cement : sand : coarse aggregate)	Group in IS 456
M5 / M7.5 / M10	1 : 5 : 10 / 1 : 4 : 8 / 1 : 3 : 6	Lean and ordinary concrete
M15	1 : 2 : 4	Ordinary (M10–M20)
M20	1 : 1.5 : 3	Ordinary. **Minimum grade for reinforced concrete**
M25	1 : 1 : 2	Standard (M25–M60) — design mix required
M65 and above	Design mix only	High strength

Durability limits for reinforced concrete (IS 456, Table 5)

- Mild exposure: minimum cement 300 kg/m³, maximum w/c 0.55, minimum grade M20.
- Moderate: 300 kg/m³, 0.50, M25. Severe: 320 kg/m³, 0.45, M30.
- Very severe: 340 kg/m³, 0.45, M35. Extreme: 360 kg/m³, 0.40, M40.
- Maximum cement content: **450 kg/m³** (more cement means more shrinkage and heat).

4. Everyday analogy

Making tea: the same spoon of tea powder in a full glass of water gives weak tea; in half a glass it gives strong tea. Cement is the tea powder and water dilutes it — that is the water–cement ratio. A plasticiser is like stirring well: you get an even, pourable mix without adding more water.

5. Confusing areas and exam traps

- **Rounded** aggregate → best workability. **Angular** aggregate → best strength. Both can be the "right" answer depending on the question.
- Abrams' law holds only for **fully compacted** concrete.
- A higher **fineness modulus** means **coarser** aggregate.
- **Flaky** is judged by thickness (0.6); **elongated** by length (1.8).
- Calcium chloride is an accelerator but is **not permitted** in reinforced concrete.
- Air entrainment improves frost resistance and workability but **reduces** strength.

6. Connections

The 0.38 water need comes from Topic 5. Workability, which the extra water buys, is measured in Topic 9. The w/c ratio is the first thing chosen in mix design (Topic 10). Exposure conditions return in the RCC module under cover and durability.

Memory code: “Less water, more strength — if compacted”

Less water, more strength	Abrams' law: strength rises as w/c falls
If compacted	5 % voids lose about 30 % strength
Round rolls, Angular anchors	Rounded → workability; angular → bond and strength
0.6 thin, 1.8 long	Flaky: thickness below 0.6 mean; elongated: length above 1.8 mean

Self-recall (close the notes, answer aloud, then check)

1. State Abrams' law.
   For fully compacted concrete with given materials, strength depends only on the water–cement ratio and falls as the ratio rises.
2. Why is calcium chloride not used in reinforced concrete?
   Chloride ions destroy the protective film on steel and cause corrosion of the reinforcement.
3. What does M25 mean?
   Concrete whose characteristic compressive strength on 150 mm cubes at 28 days is 25 N/mm².

Quick check: 5 questions

1. A concrete mix contains 400 kg of cement and 180 kg of water per cubic metre. If 20 kg more water is added on site to ease placing, the water–cement ratio changes from

   1. 0.40 to 0.45
   2. 0.45 to 0.55
   3. 0.45 to 0.50
   4. 0.50 to 0.55

1. 180/400 is 0.45, not 0.40.
2. 200/400 = 0.50.
3. Correct. 180/400 = 0.45 and 200/400 = 0.50. This small addition lowers the strength noticeably.
4. The starting ratio is 0.45.

2. **Statement:** Crushed angular aggregate is used in place of rounded river gravel, with the same proportions and water content.
Conclusion 1: The workability of the concrete decreases.
Conclusion 2: The bond between aggregate and cement paste improves.

1. Only conclusion 1 is correct
2. Only conclusion 2 is correct
3. Both conclusions 1 and 2 are correct
4. Neither conclusion 1 nor 2 is correct

1. Conclusion 2 is also true.
2. Conclusion 1 is also true.
3. Correct. Angular particles have more voids and friction (lower workability) but rougher faces and interlock (better bond, higher strength).
4. Both are true.

3. Match List-I (admixture) with List-II (purpose):

List-I	List-II
A. Calcium chloride	1. Air entrainment
B. Sugar	2. Acceleration of setting
C. Vinsol resin	3. High-range water reduction
D. Polycarboxylate ether	4. Retardation of setting

1. A-4, B-2, C-1, D-3
2. A-2, B-4, C-1, D-3
3. A-2, B-4, C-3, D-1
4. A-2, B-1, C-4, D-3

1. Calcium chloride accelerates; sugar retards.
2. Correct. Calcium chloride → accelerator; sugar → retarder; vinsol resin → air entrainment; polycarboxylate ether → superplasticiser.
3. Vinsol resin entrains air; polycarboxylate is a superplasticiser.
4. Sugar is a retarder, not an air-entraining agent.

4. In a sieve analysis of sand the cumulative percentages retained on the 4.75 mm, 2.36 mm, 1.18 mm, 600 micron, 300 micron and 150 micron sieves are 5, 15, 35, 60, 85 and 100. The fineness modulus of the sand is

1. 2.00
2. 2.50
3. 3.50
4. 3.00

1. Too low — add all six values.
2. Too low.
3. Too high.
4. Correct. (5 + 15 + 35 + 60 + 85 + 100)/100 = 300/100 = 3.00.

5. Consider the following statements:
I. As per IS 456, the minimum grade of concrete for reinforced concrete is M20.
II. Water used for mixing concrete should have a pH of not less than 6.
III. Air-entrained concrete has better resistance to freezing and thawing.
IV. A flaky particle is one whose length exceeds 1.8 times its mean dimension.

Which of the above statements are correct?

1. II and III only
2. I, II and III only
3. I, III and IV only
4. I, II, III and IV

1. Leaves out I, which is true.
2. Correct. IV describes an elongated particle; a flaky particle has thickness less than 0.6 times its mean dimension.
3. IV is false.
4. Includes the false statement IV.

Topic 9: Fresh and Hardened Concrete — Workability, Curing, Strength Tests and NDT

Syllabus line: *Concrete: workability, testing for strength, non-destructive testing.*

Asked in APPSC AEE 2023, Paper III Civil, 22 Aug Shift 2 (verified with official key): 1 question

- The **tremie pipe** is used for **underwater concreting** (Q11).

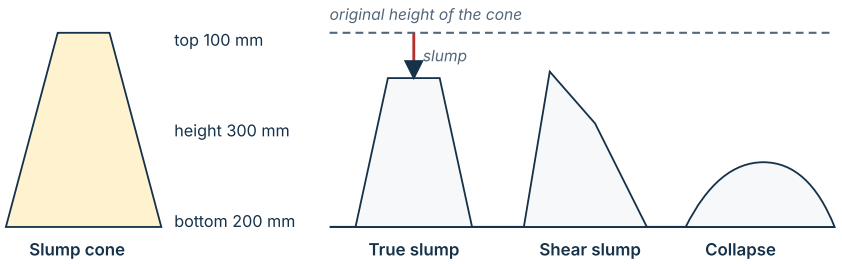
1. Basic idea

Concrete has two lives. **Fresh** concrete must be workable — easy to mix, carry, place, compact and finish without separating. **Hardened** concrete must be strong and durable. Tests exist for both. **Non-destructive tests** (NDT) judge the concrete in a finished structure without damaging it.

2. Why workability matters

Concrete that cannot be fully compacted keeps air voids, and each 1 % of voids costs about 5 % of strength. So the mix must be workable enough for the job: stiff for a road slab laid by machine, flowing for a congested column or a pile cast under water.

3. How it works



Slump is the drop in height of the concrete after the cone is lifted. A shear slump means the mix lacks cohesion — repeat the test; a collapse means the mix is too wet.

Workability test	What is measured	Best suited for
Slump test	Subsidence in mm. Cone filled in **4 layers**, each tamped **25 times** with a 16 mm rod	Medium to high workability; the usual site test
Compacting factor test	Weight of partially compacted concrete ÷ weight of fully compacted concrete (0.75 to 0.95)	Low workability (dry) mixes; more precise, a laboratory test
Vee-Bee consistometer	Time in seconds for the cone of concrete to be remoulded into a cylinder by vibration	Very stiff, dry mixes
Flow table test	Percentage increase in spread of the concrete after jolting	Very high workability, flowing concrete
Kelly ball test	Depth of penetration of a 15 cm metal ball	Quick field check on placed concrete

Degree of workability (IS 456)	Slump	Use
Very low	Measured by compacting factor 0.75–0.80	Road pavements laid by power vibrators
Low	25–75 mm	Mass concrete, lightly reinforced slabs, beams, columns, footings
Medium	50–100 mm	Heavily reinforced sections, pumped concrete
High	100–150 mm	Trench fill, in-situ piling, **tremie concrete**

Placing, compaction and curing

- **Segregation:** separation of the coarse aggregate from the mortar — caused by a badly graded or too wet mix, dropping from a height (free fall is limited to about **1.5 m**) or over-vibration.
- **Bleeding:** water rising to the surface as the solids settle. It leaves a weak, dusty top layer called **laitance**.
- **Tremie method:** for concreting **under water**. A 200–250 mm pipe with a hopper is lowered to the bottom; concrete of high slump (150–200 mm) flows down it, and the **lower end is always kept buried in the fresh concrete** so that water never mixes with it.
- **Compaction:** needle (internal) vibrator for most work; form vibrator for thin walls; surface (screed) vibrator for slabs; table vibrator for precast units.

- **Curing** keeps the concrete moist and warm so that hydration continues. IS 456: at least **7 days** for OPC and **10 days** where mineral admixtures or blended cements are used (14 days recommended in hot, dry weather). Methods: ponding, wet gunny bags, sprinkling, membrane-forming curing compound, steam curing for precast.
- **Formwork removal (OPC, normal weather):** sides of walls, columns and beams 16–24 hours; slab soffits (props left under) 3 days; beam soffits (props left) 7 days; props under slabs — 7 days up to 4.5 m span, 14 days above; props under beams — 14 days up to 6 m span, 21 days above.

Strength tests on hardened concrete (IS 516)

- **Cube test:** 150 mm cubes, filled in 3 layers with 35 strokes each (or vibrated), cured in water, tested at **7 and 28 days**, load applied at **14 N/mm² per minute**. Strength = failure load ÷ area. The average of three cubes is one result; individual values must be within ±15 % of the average.
- **Cylinder test:** 150 mm diameter × 300 mm. Cylinder strength ≈ **0.8 × cube strength**.
- **Gain of strength:** about 40 % at 3 days, **65–70 % at 7 days**, 100 % (by definition) at 28 days.
- **Flexural strength** (modulus of rupture): beam 150 × 150 × 700 mm under two-point loading. IS 456 estimate: $\mathbf{f_{cr} = 0.7\sqrt{f_{ck}}}$.
- **Split tensile strength:** a cylinder loaded along its side splits; f_{ct} = **$\mathbf{2P \div (\pi D L)}$**. Tensile strength is only about 8–12 % of compressive strength.
- **Modulus of elasticity:** E_c = **$\mathbf{5000\sqrt{f_{ck}}}$** N/mm².
- **Characteristic strength** f_{ck}: the strength below which not more than **5 %** of test results are expected to fall.

Non-destructive test	Principle	What it tells
Rebound (Schmidt) hammer — IS 13311 Part 2	A spring-driven mass strikes the surface; the **rebound number** measures surface hardness	Approximate strength and uniformity of the surface layer. Affected by surface smoothness, moisture and carbonation
Ultrasonic pulse velocity (UPV) — IS 13311 Part 1	Time of travel of an ultrasonic pulse between two transducers; velocity = path length ÷ time. **Direct** transmission (opposite faces) is the best	Quality and homogeneity, cracks, voids, honeycombing. Above 4.5 km/s excellent; 3.5–4.5 good; 3.0–3.5 medium; below 3.0 doubtful
Penetration resistance (Windsor probe)	Depth to which a steel probe is fired into the concrete	Strength near the surface
Pull-out test	Force needed to pull out a cast-in or drilled-in insert	In-place strength
Core test (partly destructive)	A cylinder is drilled out and crushed	The most reliable in-place strength. Acceptable if the average core strength is at least **85 %** of the cube strength of the grade and no core is below **75 %**
Cover meter (pachometer)	Electromagnetic	Position, cover and size of bars
Half-cell potential	Electrical potential of the steel	Likelihood of corrosion of reinforcement
Carbonation test	**Phenolphthalein** sprayed on a fresh break: pink = still alkaline; colourless = carbonated	Depth of carbonation

4. Everyday analogy

Workability is like dosa batter and idli batter: each is right for its own job, and the wrong consistency spoils the result. Curing is like watering a newly planted sapling for the first weeks — skip it and growth stops, however good the seed. The rebound hammer is like tapping a coconut or a pot to judge it without breaking it.

5. Confusing areas and exam traps

- Slump test → medium and high workability; compacting factor → low; Vee-Bee → very stiff mixes; flow table → flowing mixes.
- Slump cone is filled in **4** layers of 25 strokes; the cube mould in **3** layers of 35 strokes.
- Cylinder strength is **lower** than cube strength (about 0.8 times).
- The rebound hammer measures **surface hardness**; UPV looks at the **interior**.
- In tremie work the pipe end stays **inside** the concrete; the mix needs **high** slump and is not vibrated.
- Bleeding is a form of segregation in which **water** separates; laitance is its result.

6. Connections

f_{ck}, 0.7 $\sqrt{f_{ck}}$ and 5000 $\sqrt{f_{ck}}$ are used throughout the RCC module. The split test relies on the lateral tension explained for masonry in Topic 4. Tremie concreting returns with piles and well foundations in the Foundation Engineering module.

Memory code: "S-C-V-F: wet to dry is S-F reversed" and "4 × 25, 3 × 35"

S — Slump	Medium to high workability
C — Compacting factor	Low workability
V — Vee-Bee	Very stiff mixes (time in seconds)
F — Flow table	Flowing mixes
4 × 25, 3 × 35	Slump cone: 4 layers, 25 strokes. Cube: 3 layers, 35 strokes
7 – 10	Curing: 7 days OPC, 10 days blended cements

Self-recall (close the notes, answer aloud, then check)

1. What is the tremie method and its one essential rule?
   Placing concrete under water through a vertical pipe; the lower end of the pipe must always remain buried in the fresh concrete.
2. Give the IS 456 expressions for flexural strength and modulus of elasticity of concrete.
   $f_{cr} = 0.7\sqrt{f_{ck}}$ and $E_c = 5000\sqrt{f_{ck}}$ (N/mm²).
3. Which NDT method detects internal voids and cracks, and which measures surface hardness?
   Ultrasonic pulse velocity detects internal flaws; the rebound hammer measures surface hardness.

Quick check: 5 questions

1. A 150 mm concrete cube fails at a load of 675 kN. Its compressive strength is

   1. 4.5 N/mm²
   2. 45 N/mm²
   3. 22.5 N/mm²
   4. 30 N/mm²

   1. This divides by the side, not the area.
   2. This divides by 15,000.
   3. This divides by 30,000.
   4. Correct. Area = 150 × 150 = 22,500 mm²; 675,000/22,500 = 30 N/mm².

2. Match List-I (test) with List-II (what is measured):

List-I	List-II
A. Vee-Bee test	1. Ratio of weights of partially and fully compacted concrete
B. Compacting factor test	2. Time of travel of an ultrasonic pulse
C. Rebound hammer test	3. Time in seconds for remoulding
D. UPV test	4. Surface hardness

   1. A-3, B-1, C-2, D-4
   2. A-1, B-3, C-4, D-2
   3. A-3, B-1, C-4, D-2
   4. A-4, B-1, C-3, D-2

   1. Rebound hammer → surface hardness; UPV → pulse time.
   2. Vee-Bee gives a time; compacting factor gives a ratio.
   3. Correct. Vee-Bee → time in seconds; compacting factor → ratio of weights; rebound hammer → surface hardness; UPV → transit time of a pulse.
   4. Vee-Bee does not measure hardness.

3. **S1:** In the tremie method, the lower end of the pipe is kept above the surface of the fresh concrete so that the concrete falls freely through the water.
S2: Concrete placed by tremie should have a high slump.

1. Both S1 and S2 are true
2. S1 is true and S2 is false
3. S1 is false and S2 is true
4. Both S1 and S2 are false

1. S1 is false.
2. S1 is false; S2 is the true one.
3. Correct. The pipe end must stay buried in the fresh concrete so that water does not wash out the cement. A slump of about 150–200 mm is needed because the concrete cannot be vibrated.
4. S2 is true.

4. The modulus of rupture of M25 concrete, as estimated by IS 456, is

1. 2.5 N/mm²
2. 3.5 N/mm²
3. 5.0 N/mm²
4. 17.5 N/mm²

1. This is 10 % of 25.
2. Correct. 0.7 √25 = 0.7 × 5 = 3.5 N/mm².
3. This is √25 alone.
4. This is 0.7 × 25.

5. Consider the following statements:
I. Bleeding of concrete produces a weak layer of laitance at the surface.
II. The cylinder strength of concrete is greater than its cube strength.
III. IS 456 requires a minimum curing period of 7 days for concrete made with ordinary Portland cement.
IV. A core test is more reliable than the rebound hammer test for in-place strength.
Which of the above statements are correct?

1. I, III and IV only
2. II, III and IV only
3. III and IV only
4. I and III only

1. Correct. II is false: cylinder strength is about 0.8 times cube strength.
2. II is false.
3. Leaves out I, which is true.
4. Leaves out IV, which is true.

Topic 10: Concrete Mix Design (IS 10262) — and Reinforcing Steel

Syllabus line: *Concrete: mix design methods.* Steel is not named in the Building Materials syllabus line but two questions came from it in 2023, so it is added here.

Asked in APPSC AEE 2023, Paper III Civil, 22 Aug Shift 2 (verified with official key): 2 questions (both on steel)

- As per **IS 1786 : 2008**, the yield strength of high-strength deformed bars is taken as the **0.2 % proof stress** (Q16).
- Two statements on carbon in steel: S1 said that ductility increases with carbon — **false** (it decreases). S2 said that the modulus of elasticity does not depend on carbon content — **true** (Q17).

1. Basic idea

Mix design is choosing the proportions of cement, water, fine and coarse aggregate (and admixture) so that the concrete has the required **workability** when fresh, the required **strength** and **durability** when hard, at the **lowest cost**. A **nominal mix** uses fixed volume proportions such as 1 : 1.5 : 3; a **design mix** is worked out for the actual materials and is required for M25 and above.

2. Why design for more than the grade

Cube results scatter round their average. If the average were only equal to the grade, half the results would fall below it. So the mix is designed for a higher **target mean strength**, chosen so that only 5 % of results fall below the grade.

3. How it works

Target mean strength

- **$f'_{ck} = f_{ck} + 1.65\ s$**, where s is the standard deviation. (IS 10262 : 2019 also asks for f_{ck} + X and takes the larger; X is 5 for M10–M15, 5.5 for M20–M25, 6.5 for M30–M60.)
- The factor **1.65** corresponds to **5 %** of results being allowed below f_{ck} (1 in 20).
- Assumed standard deviation (IS 456 / IS 10262) when no site data exist: **3.5** N/mm² for M10–M15; **4.0** for M20–M25; **5.0** for M30–M60.
- Example: M25 → 25 + 1.65 × 4 = **31.6 N/mm²**. M30 → 30 + 1.65 × 5 = 38.25 N/mm².
- Good quality control lowers s, lowers the target strength and so saves cement.

Step (IS 10262)	What is done
1. Target strength	$f'_{ck} = f_{ck} + 1.65\ s$
2. Water–cement ratio	Chosen from the strength–w/c curve for the cement used; must not exceed the durability limit of IS 456
3. Water content	From a table, by maximum size of aggregate, for 50 mm slump: about **208 kg/m³** for 10 mm, **186** for 20 mm, **165** for 40 mm aggregate. Add about 3 % for every extra 25 mm of slump; reduce when a plasticiser is used
4. Cement content	Water ÷ w/c ratio. Check against the minimum for durability and the maximum of 450 kg/m³
5. Coarse aggregate fraction	Volume of coarse aggregate per unit volume of total aggregate, from a table by **zone of sand** and size of aggregate (for 20 mm aggregate at w/c 0.5: Zone I 0.60, II 0.62, III 0.64, IV 0.66). Finer sand → more coarse aggregate
6. Quantities	By the **absolute volume method**: 1 m³ = volumes of cement + water + admixture + air + aggregates, each = mass ÷ (specific gravity × 1000)
7. Site corrections	Adjust for surface moisture and absorption of the aggregates
8. Trial mixes	Cast, test and adjust until workability and strength are satisfied

Other methods of mix design: ACI committee 211 method (USA), DOE method (Britain), Road Note No. 4 method, fineness modulus method, maximum density method, and the arbitrary (nominal) method. The **IS method (IS 10262)** is the one used in India.

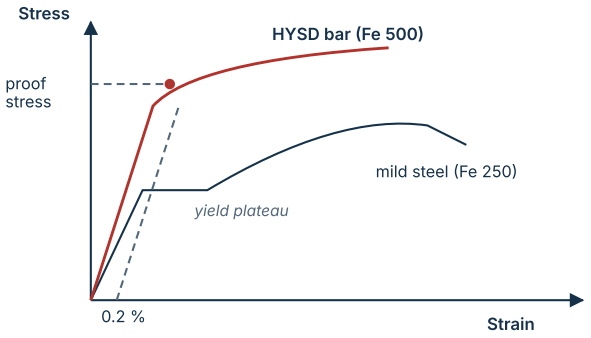
Reinforcing and structural steel (asked in 2023)

Material	Carbon content	Character
Wrought iron	Below about 0.15 %	Soft, ductile, fibrous; forged and welded easily
Mild steel (low carbon)	About 0.15–0.30 %	Ductile, weldable, definite yield point. Structural steel and plain bars
Medium carbon steel	0.30–0.60 %	Rails, axles
High carbon steel	0.60–1.5 %	Hard, strong, less ductile. Tools, springs, prestressing wires
Cast iron	2–4 %	Hard, brittle, strong in compression only; cannot be forged

Effect of carbon and the IS 1786 bars

- More carbon → **higher** tensile strength, yield strength, hardness and brittleness; **lower** ductility, toughness and weldability.
- The **modulus of elasticity** of all steels is practically the same, **2×10^5 N/mm²**, whatever the carbon content.
- **IS 432** — mild steel plain bars, Fe 250. **IS 1786 : 2008** — high-strength deformed bars: **Fe 415, Fe 500, Fe 550, Fe 600**, with ductile variants Fe 415D, Fe 500D, Fe 550D and Fe 415S, Fe 500S. **IS 2062** — structural steel sections (E 250 and above).
- Mild steel shows a clear yield point. High-strength bars do not, so their yield strength is defined as the **0.2 % proof stress**: the stress at which a line drawn parallel to the initial straight part, from 0.2 % strain, meets the curve.
- Minimum elongation: Fe 415 — 14.5 %; Fe 500 — 12 %; Fe 500D — 16 %; Fe 550 — 10 % (mild steel about 23 %). Higher grade, lower ductility.

- **TMT bars** (thermo-mechanically treated): the hot bar is quenched with water so that the surface becomes hard martensite while the core stays soft and ductile — high strength with good ductility and weldability.
- Ribs (deformations) on the bar improve the **bond** with concrete; IS 456 allows 60 % higher bond stress for deformed bars.



0.2 % proof stress

1. Mark a strain of 0.002 on the strain axis.
2. Draw a line parallel to the initial straight part.
3. Where it cuts the curve is the proof stress.

Both curves start with the same slope: E is the same for all steels (2×10^5 N/mm²).

The stronger bar breaks at a smaller strain: more strength, less ductility.

Mild steel has a visible yield plateau; a high-strength bar does not, so IS 1786 defines its yield strength as the 0.2 % proof stress. The two curves share the same initial slope.

4. Everyday analogy

A tailor who must deliver trousers of at least 100 cm length does not aim at exactly 100, because his cutting varies a little; he aims at 102 so that almost none comes out short. That margin is 1.65 s. A careful tailor (small s) wastes less cloth. For carbon in steel, think of jaggery syrup: cook it longer and the sweet becomes harder and stronger to bite, but it snaps instead of bending.

5. Confusing areas and exam traps

- Target strength is f_{ck} **+ 1.65 s**, not f_{ck} + s. The 1.65 belongs to 5 % defectives.
- Water content depends on the **size of aggregate and slump**; cement content then follows from the w/c ratio — not the other way round.
- Finer sand (Zone IV) → **larger** proportion of coarse aggregate.
- Carbon raises strength and hardness but **lowers ductility**; it does **not** change E. This was the 2023 trap.
- Fe 500 means a yield (0.2 % proof) strength of 500 N/mm², not the ultimate strength.

6. Connections

Every earlier topic feeds this one: cement strength (Topic 6), sand zone and aggregate size (Topic 8), w/c ratio and durability limits (Topic 8), slump (Topic 9). The stress–strain curve of steel was drawn in Module 1 (Strength of Materials) and governs the design stress 0.87 f_y in the RCC module.

Memory code: "Target = grade + 1.65 s" and "Carbon up: strong, hard, brittle — E same"

1.65 → 5 %	Only 5 % of results may fall below the grade
3.5 – 4 – 5	Standard deviation: M10–15, M20–25, M30–60
186 for 20	Water content for 20 mm aggregate at 50 mm slump
Carbon up	Strength and hardness up; ductility and weldability down; E unchanged
0.2 % proof	Yield strength of HYSD bars in IS 1786

Self-recall (close the notes, answer aloud, then check)

1. Write the target mean strength of M20 concrete.
   20 + 1.65 × 4 = 26.6 N/mm².
2. List the first four steps of the IS 10262 method.
   Target mean strength; water–cement ratio; water content; cement content (then aggregate proportions and trial mixes).
3. How is the yield strength of an HYSD bar defined, and why?
   As the 0.2 % proof stress, because such bars have no definite yield point.

Quick check: 5 questions

1. The target mean strength for M30 concrete, taking the standard deviation as 5 N/mm², is

   1. 35.00 N/mm²
   2. 36.60 N/mm²
   3. 31.65 N/mm²
   4. 38.25 N/mm²

   1. This adds s only once.
   2. This uses s = 4, the value for M20–M25.
   3. This adds 1.65 without multiplying by s.
   4. Correct. 30 + 1.65 × 5 = 38.25 N/mm².

2. In a mix design the water content is fixed at 180 kg/m³ and the water–cement ratio at 0.45. The durability requirement is a minimum cement content of 320 kg/m³. The cement content to be adopted is

   1. 320 kg/m³
   2. 400 kg/m³
   3. 450 kg/m³
   4. 81 kg/m³

   1. The minimum is only a lower check; the mix needs more.
   2. Correct. Cement = 180/0.45 = 400 kg/m³, which is above the minimum of 320 and below the maximum of 450.
   3. 450 kg/m³ is the upper limit, not the design value.
   4. This multiplies instead of dividing.

3. **S1:** With an increase in carbon content, the ductility of steel increases.
   S2: The modulus of elasticity of steel is practically independent of its carbon content.

   1. Both S1 and S2 are true
   2. S1 is true and S2 is false
   3. S1 is false and S2 is true
   4. Both S1 and S2 are false

   1. S1 is false.
   2. S1 is false.
   3. Correct. Carbon makes steel stronger and harder but less ductile; E stays at about 2 × 10⁵ N/mm².
   4. S2 is true.

4. **Assertion (A):** The yield strength of high-strength deformed bars is specified as the 0.2 % proof stress.
   Reason (R): Such bars do not show a definite yield point in the tension test.

   1. Both A and R are true, and R is the correct explanation of A
   2. Both A and R are true, but R is NOT the correct explanation of A
   3. A is true, but R is false
   4. A is false, but R is true

   1. Correct. With no visible yield plateau, a conventional offset definition is needed.
   2. R is the reason.
   3. R is true.
   4. A is true.

5. Consider the following statements about concrete mix design by IS 10262:

   I. The target mean strength is higher than the characteristic strength.
   II. Water content is selected on the basis of the maximum size of aggregate and the required slump.
   III. For a finer zone of sand, the proportion of coarse aggregate is reduced.
   IV. Final proportions are confirmed by trial mixes.

   Which of the above statements are correct?

   1. I and II only
   2. II, III and IV only
   3. II and IV only
   4. I, II and IV only

   1. Leaves out IV, which is true.
   2. III is false.
   3. Leaves out I, which is true.
   4. Correct. III is false: finer sand (Zone IV) allows a larger proportion of coarse aggregate.

Notebook page — write these from memory

Close this page. In your notebook, write the items below in this order without looking. Then open the page and correct yourself in a different colour. Whatever you could not write is your revision list for Sunday.

Topic 1

1. Sketch of the trunk cross-section with five labels
2. Softwood vs hardwood; six species with one use each
3. Moisture content formula; fibre saturation point; what happens above and below it
4. Shrinkage order T-R-L; strength in different directions; three moisture methods of IS 11215

Topic 2

1. Three shakes sketch (heart, star, cup)
2. Defects by source: natural, seasoning, conversion, fungi, insects — three of each
3. Objects of seasoning; air vs kiln seasoning
4. Preservatives: creosote, ASCU 1 : 3 : 4; Bethell vs Lowry–Rueping; Abel
5. Plywood: odd plies, crossed grain, four advantages

Topic 3

1. Five ingredients of brick earth with proportion and job
2. Four stages of manufacture; frog; modular size and 500 bricks per m^3
3. Defects: chuffs, bloating, black core, efflorescence, lime blowing
4. Traditional classes; IS 1077 classes; absorption 20 % and 15 %; saturation coefficient

Topic 4

1. Why masonry is weaker than brick (lateral tension)
2. Seven factors affecting masonry strength; effect of mortar strength
3. Mortar types H1 to M3; slenderness ratio limit 27
4. Stones: three classifications; metamorphic pairs; M-I-A-B tests and Smith's test

Topic 5

1. Manufacture in one line; role of gypsum; oxide composition
2. Table of four Bogue compounds: share, role, heat
3. Hydration products; 38 % = 23 % + 15 %
4. Ten types of cement: what is changed and where used; false set vs flash set

Topic 6

1. Vicat sketch with three attachments
2. Limits: fineness, initial and final set, soundness (two tests)
3. Water rules: P, 0.85 P, 0.78 P, (P/4 + 3) %
4. Strengths of 33, 43, 53 grades at 3, 7, 28 days; 1440 kg/m^3; storage loss

Topic 7

1. Functions of sand; table of proportions for eight works
2. Water demand and water retentivity; gauged mortar
3. Bulking graph; dry volume = 1.3 × wet; one worked cement quantity
4. Plaster coats, thicknesses, defects; lime: calcination, slaking, setting; fat vs hydraulic

Topic 8

1. Abrams' law with graph; effect of voids
2. Aggregate shapes; flaky 0.6, elongated 1.8; fineness modulus; limits 30 % and 45 %
3. Water limits (pH, chlorides); six admixtures with one example
4. Nominal mixes M5 to M25; durability table; maximum cement 450

Topic 9

1. Slump cone dimensions; three slumps; four workability tests and their range
2. Segregation, bleeding, laitance; tremie rule; curing 7 and 10 days
3. Cube test details; 0.8; 0.7 √fck; 2P/πDL; 5000 √fck
4. NDT table: rebound hammer, UPV grades, core 85 % and 75 %, carbonation test

Topic 10

1. Target strength formula; 1.65 and 5 %; standard deviations
2. Eight steps of IS 10262 with 186 kg for 20 mm
3. Carbon table from wrought iron to cast iron; effect of carbon; E
4. IS 1786 grades; 0.2 % proof stress sketch; TMT bars

References

- APPSC Notification No. 27/2026 (Assistant Executive Engineers) — syllabus of Paper 3 Civil, item 1 (Building Materials).
- APPSC AEE Paper III (Civil Engineering), held on 22 Aug 2023, Shift 2 — official paper with key. All "Asked in APPSC AEE 2023" notes refer to this paper.
- Indian Standards: IS 456 : 2000 (plain and reinforced concrete), IS 10262 : 2019 (mix proportioning), IS 383 (aggregates), IS 269 and IS 4031 (cement and its tests), IS 1077 and IS 3495 (bricks), IS 1905 (masonry), IS 2250 (mortar), IS 712 (lime), IS 883 and IS 11215 (timber), IS 516 and IS 13311 (tests on concrete), IS 1786 : 2008 (reinforcing bars).
- Textbooks: S.C. Rangwala, *Engineering Materials*; S.K. Duggal, *Building Materials*; M.S. Shetty, *Concrete Technology*. Where textbooks give slightly different figures (for example the proportions of brick earth), the page says so and shows the value used in the 2023 key.
- Photos: Wikimedia Commons, free licences; each photo links to its source page.
- Quiz questions are original practice questions (Sulabham), not previous-year questions.

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